Extensions to the `ltxdoc` class *

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This file embodies the `ltxdocext` package, the implementation and its user documentation.

The distribution point for this work is ftp://ftp.teleport.com/users/ogawa/macros/latex/contrib/supported/ltxdocext..., which contains fully unpacked, prebuilt runtime files and documentation.

To use this document class, you must have a working TEX installation equipped with LATEX 2_ε and possibly pdftex and Adobe Acrobat Reader or equivalent.

To install, retrieve the distribution, unpack it into a directory on the target computer, and move the files `ltxdocext.sty` and `acrofont.sty` into a location in your filesystem where they will be found by LATEX.

If you will be using the `acrofont` package, you must also install the virtual fonts `zpsynocmrv`, `zptmnocmr`, `zptmnocmrm`, and `zpzcnocmry`. The corresponding `.tfm`, `.vf`, and `.vpl` files are part of this distribution.

To use, read the user documentation `ltxdocext.pdf`. The `.dtx` file, `ltxdocext.dtx`, constitutes in itself an instance of use of the `ltxdocext` package and the `acrofont` package.

Contents

1 Processing Instructions **2**
1.1 Build Instructions . . . 3
1.2 Bill of Materials . . . 3
1.2.1 Primary Source . . . 3
1.2.2 Generated by `latex ltxutil.dtx` . . . 3
1.2.3 Generated by `tex ltxutil.ins` . . . 3
1.2.4 Documentation . . . 4
1.2.5 Auxiliary . . . 4

2 Code common to all modules **4**

*This file has version number 1.0d, last revised 2020/09/30. For version number and date, search on "1.0d" in the .dtx file, or see the end of the 00readme.txt file.

3 The driver module doc **4**
3.1 The Preamble . . . 5
3.1.1 Docstrip and info directives . . . 5
3.2 The installer file . . . 5
3.3 The "Read Me" File . . . 6
3.4 The Document Body . . . 8

4 Using this package **8**
4.1 Invoking the package . . . 9

5 Compatability with LaTeX's Required Packages **9**
5.1 array . . . 9
5.2 longtable . . . 10

6 Implementation of package **10**
6.1 Beginning of the ltxutil DOCSTRIP module . . . 10
6.2 Banner . . . 10
6.3 Errors and warnings . . . 10
6.4 New Tools . . . 11
6.5 Boolean Control . . . 12
6.6 Begin Document Structure . . . 15
6.7 Type Tools . . . 18
6.8 Display Math . . . 18
6.9 Footnotes . . . 20
6.10 Floats . . . 23
6.10.1 Usage notes . . . 23
6.10.2 Robustifying fragile commands . . . 24
6.10.3 Preparing for the hyperref package . . . 24
6.10.4 Footnotes within floats, unfloating floats, float font . . . 25
6.10.5 Writing floats out to a file . . . 27
6.11 Counters . . . 30
6.12 Customization of Sections . . . 30
6.13 Patch the tabular and array Environments . . . 34
6.14 Repair other broken parts of LaTeX . . . 55
6.15 Syntax . . . 56
6.16 Auto-indented Contents . . . 56
6.17 Lists . . . 59
6.18 End of the ltxutil DOCSTRIP module . . . 60

Index **61**

1 Processing Instructions

The package files ltxdocext.sty and acrofont.sty are generated from this file, ltxdocext.dtx, via the DOCSTRIP facility of LaTeXvia tex ltxdocext.ins. The typeset documentation that you are now reading is generated from the

same file by typesetting it with LaTeX or pdftex via `latex ltxdocext.dtx` or `pdflatex ltxdocext.dtx`.

1.1 Build Instructions

You may bootstrap this suite of files solely from `ltxdocext.dtx`. Prepare by installing LaTeX 2_ε (and either tex or pdftex) on your computer, then carry out the following steps:

1. Within an otherwise empty directory, typeset `ltxdocext.dtx` with LaTeX or pdflatex; you will obtain the typeset documentation you are now reading, along with the installer `ltxdocext.ins`, and the file `00readme.txt`.

2. Now typeset `ltxdocext.ins`, thereby generating the package file `ltxdocext.sty`, and the package file `acrofont.sty`. Make sure that DOCSTRIP receives permission to overwrite existing versions of these packages.

3. Install `ltxdocext.sty` and `acrofont.sty` by moving them to a location in your filesystem where they will be found by LaTeX.

4. Now complete the typesetting of the documentation by retypesetting `ltxdocext.dtx`. Note: you will have to run LaTeX twice, then `makeindex`, then LaTeX again in order to obtain a valid index and table of contents.

1.2 Bill of Materials

Following is a list of the files in this distribution arranged according to provenance.

1.2.1 Primary Source

One single file generates all.

```
%ltxutil.dtx
%
```

1.2.2 Generated by `latex ltxutil.dtx`

Typesetting the source file under LaTeX generates the readme and the installer.

```
%00readme.txt   ltxutil.ins
%
```

1.2.3 Generated by `tex ltxutil.ins`

Typesetting the installer generates the package files.

```
%ltxutil.sty
%
```

1.2.4 Documentation

The following are the online documentation:

```
%ltxutil.pdf
%
```

1.2.5 Auxiliary

The following are auxiliary files generated in the course of running LaTeX:

```
%ltxutil.aux ltxutil.idx ltxutil.ind ltxutil.log ltxutil.toc
%
```

2 Code common to all modules

The following may look a bit klootchy, but we want to require only one place in this file where the version number is stated, and we also want to ensure that the version number is embedded into every generated file.

Now we declare that these files can only be used with LaTeX 2_ε. An appropriate message is displayed if a different TeX format is used.

```
%<*doc|ltxutil>
\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%</doc|ltxutil>
```

As desired, the following modules all take common version information:

```
%<ltxutil>\ProvidesFile{ltxutil.sty}%
%<*doc>
\expandafter\ProvidesFile\expandafter{\jobname.dtx}%
%</doc>
```

The following line contains, for once and for all, the version and date information. By various means, this information is reproduced consistently in all generated files and in the typeset documentation.

```
%<*doc|ltxutil>
 [2020/09/30 1.0d utilities package]% \fileversion
%</doc|ltxutil>
```

3 The driver module `doc`

This module, consisting of the present section, typesets the programmer's documentation, generating the `.ins` installer and `00readme.txt` as required.

Because the only uncommented-out lines of code at the beginning of this file constitute the `doc` module itself, we can simply typeset the `.dtx` file directly, and there is thus rarely any need to generate the "doc" DOCSTRIP module. Module delimiters are nonetheless required so that this code does not find its way into the other modules.

The `\end{document}` command concludes the typesetting run.

```
%<*doc>
```

3.1 The Preamble

The programmers documentation is formatted with the `ltxdoc` class with local customizations, and with the usual code line indexing.

```
\documentclass{ltxdoc}
\RequirePackage{ltxdocext}%
\let\url\undefined
\RequirePackage[colorlinks=true,linkcolor=blue]{hyperref}%
\pdfstringdefDisableCommands{%
  \let\file\relax
  \let\sc\relax
}
\makeatletter
\@ifundefined{package@font}{}%
  {\expandafter\RequirePackage\expandafter{\csname package@font\endcsname}}
\makeatother
\CodelineIndex\EnableCrossrefs
```

3.1.1 Docstrip and info directives

We use so many DOCSTRIP modules that we set the `StandardModuleDepth` counter to 1.

```
\setcounter{StandardModuleDepth}{1}
```

The following command retrieves the date and version information from this file.

```
\expandafter\GetFileInfo\expandafter{\jobname.dtx}%
```

3.2 The installer file

The installer `ltxutil.ins` appears here. If you have retrieved the standard distribution of this package, the installer file is already on your filesystem. If you are bootstrapping, the first typesetting of the `.dtx` file will cause the installer to be generated.

The following modules are used to direct DOCSTRIP in generating the external files:

Module	File	Description
doc	`ltxutil.drv`	driver for programmer's documentation
ltxutil,ltxutil-krn	`ltxutil.sty`	this package
ltxutil-krn		the portion of this package suitable for inclusion within another pac

```
\begin{filecontents}{ltxutil.ins}
%% This file will generate documentation and runtime files
%% from ltxutil.dtx when run through LaTeX or TeX.
\input docstrip
\preamble

This is a generated file;
```

```
34 altering it directly is inadvisable;
35 instead, modify the original source file.
36 See the URL in the file 00readme.txt.
37
38 Copyright notice.
39
40    These files are distributed
41    WITHOUT ANY WARRANTY; without even the implied warranty of
42    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
43
44 \endpreamble
45 \keepsilent
46   \generate{%
47    \file{ltxutil.drv}{\from{ltxutil.dtx}{doc}}%
48    \file{ltxutil.sty}{%
49     \from{ltxutil.dtx}{ltxutil,ltxutil-krn}%
50    }%
51   }%
52 \ifToplevel{
53 \Msg{*************************************************************}
54 \Msg{*}
55 \Msg{* To finish the installation, please move}
56 \Msg{*    ltxutil.sty}
57 \Msg{* into a directory searched by TeX.}
58 \Msg{*}
59 \Msg{* To produce the documentation,
60        run ltxutil.dtx through LaTeX.}
61 \Msg{*}
62 \Msg{* Happy TeXing}
63 \Msg{*************************************************************}
64 }
65 \endbatchfile
66 \end{filecontents}
```

Note that, because all of the files generated by the installer are part of the standard distribution, it will be necessary to run the installer only when bootstrapping (or, of course, during development). Note, too, that it is rare to generate the `doc` module because it suffices to simply typeset the `.dtx` file itself.

3.3 The "Read Me" File

As promised above, here is the contents of the "Read Me" file. That file serves a double purpose, since it also constitutes the beginining of the programmer's documentation. What better thing, after all, to have appear at the beginning of the typeset documentation?

A good discussion of how to write a ReadMe file can be found in Engst, Tonya, "Writing a ReadMe File? Read This" *MacTech* October 1998, p. 58.

Note the appearance of the `\StopEventually` command, which marks the dividing line between the user documentation and the programmer documentation.

The usual user will not be asked to do a full build, not to speak of the bootstrap. Instructions for carrying these processes begin the programmer's manual.

```
\begin{filecontents*}{00readme.txt}
\title{%
 A \LaTeX\ Package of utility macros%
 \thanks{%
  This file has version number \fileversion,
  last revised \filedate.%
  % For version number and date,
  % search on "\fileversion" in the .dtx file,
  % or see the end of the 00readme.txt file.
 }%
}%

\author{%
Arthur Ogawa (\texttt{mailto:ogawa@teleport.com}),
\fileversion\\Copyright (C) 1999 Arthur Ogawa
}%
\maketitle

This file embodies the \classname{ltxutil} package,
the implementation and its user documentation.

The distribution point for this work is
\url{ftp://ftp.teleport.com/users/ogawa/macros/latex/contrib/supported/ltxutil...},
which contains fully unpacked, prebuilt runtime files and documentation.

The \classname{ltxutil} package was commissioned by the American Physical Society
and is distributed under the terms of the \LaTeX\ Project Public License,
the same license under which all the portions of \LaTeX\ itself is distributed.
Please see \url{http://ctan.tug.org/macros/latex/base/lppl.txt} for details.

To use this document class, you must have a working
\TeX\ installation equipped with \LaTeXe\
and possibly pdftex and Adobe Acrobat Reader or equivalent.

To install, retrieve the distribution,
unpack it into a directory on the target computer,
and move the file \file{ltxutil.sty}
into a location in your filesystem where it will be found by \LaTeX.

To use, read the user documentation \file{ltxutil.pdf}.

\tableofcontents

\section{Processing Instructions}

The package file \file{ltxutil.sty}
is generated from this file, \file{ltxutil.dtx},
```

```
using the {\sc docstrip} facility of \LaTeX
via |tex ltxutil.ins|.
The typeset documentation that you are now reading is generated from
the same file by typesetting it with \LaTeX\ or pdftex
via |latex ltxutil.dtx| or |pdflatex ltxutil.dtx|.

\subsection{Build Instructions}

You may bootstrap this suite of files solely from \file{ltxutil.dtx}.
Prepare by installing \LaTeXe\ (and either tex or pdftex) on your computer,
then carry out the following steps:
\begin{enumerate}
\item
Within an otherwise empty directory,
typeset \file{ltxutil.dtx} with \LaTeX\ or pdflatex;
you will obtain the typeset documentation you are now reading,
along with
the installer \file{ltxutil.ins},
and the file \file{00readme.txt}.

Note: you will have to run \LaTeX\ twice, then \file{makeindex}, then
\LaTeX\ again in order to obtain a valid index and table of contents.
\item
Now typeset \file{ltxutil.ins},
thereby generating the package file \file{ltxutil.sty}.
\item
Install \classname{ltxutil.sty}
by moving it to a location
in your filesystem where they will be found by \LaTeX.
\end{enumerate}
\end{filecontents*}
```

3.4 The Document Body

Here is the document body, containing only a `\DocInput` directive—referring to this very file. This very cute self-reference is a common `ltxdoc` idiom.

```
\begin{document}%
\expandafter\DocInput\expandafter{\jobname.dtx}%
% ^^A\PrintChanges
\end{document}

%</doc>
```

4 Using this package

Once this package is installed on your filesystem, you can employ it in adding functionality to LaTeX by invoking it in your document or document class.

4.1 Invoking the package

In your document, you can simply call it up in your preamble:

```
%\documentclass{book}%
%\usepackage{ltxutil}%
%\begin{document}
%⟨your document here⟩
%\end{document}
%
```

However, the preferred way is to invoke this package from within your customized document class:

```
%\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%\ProvidesClass{myclass}%
%\RequirePackage{ltxutil}%
%\LoadClass{book}%
%⟨class customization commands⟩
%\endinput
%
```

Once loaded, the package gives you acccess to certain procedures, usually to be invoked by a LaTeX command or environment, but not at the document level.

5 Compatability with LaTeX's Required Packages

Certain packages, usually ones written by members of the LaTeX Project itself, have been designated "required" and are distributed as part of standard LaTeX. These packages have been placed in a priviledged position vis á vis the LaTeX kernel in that they override the definitions of certain kernel macros.

The `ltxutil` package will be incompatible with any package that redefines any of the kernel macros that `ltxutil` patches—if that package is loaded *after* `ltxutil`. This means that for greatest compatability, `ltxutil` should be loaded *after*, say, `ftnright`, which overwrites LaTeX's kernel procedures `\@outputdblcol`, `\@startcolumn`, and `\@makecol`.

Hereinafter follows some notes on specific LaTeX packages.

5.1 array

This package alters the way tabular environments are done, therefore it could run afoul of the LaTeX "required" package `array` or any package that calls for it to be loaded. However, this package has provisions for remaining compatible with `array`. So long as the version of `array` that is used with this package has the appropriate meanings for the procedures it overwrites, all should be well.

5.2 longtable

David Carlisle's `longtable` package modifies both the LaTeX kernel and the `array` package. This package must therefore alter `\LT@array`. For now, that job is handled by `ltxgrid`.

6 Implementation of package

Special acknowledgment: this package uses concepts pioneered and first realized by William Baxter (mailto:web@superscript.com) in his SuperScript line of commercial typesetting tools, and which are used here with his permission.

6.1 Beginning of the `ltxutil` DOCSTRIP module

```
%<*ltxutil>
\def\package@name{ltxutil}%
\expandafter\PackageInfo\expandafter{\package@name}{%
 Utility macros for \protect\LaTeXe,
 by A. Ogawa (ogawa@teleport.com)%
}%
%</ltxutil>
```

6.2 Banner

Credit where due.

```
%<*ltxutil-krn>
\typeout{%
 ltxutil: portions licensed from W. E. Baxter (web@superscript.com)%
}%
```

6.3 Errors and warnings

`\class@err` `\class@warn` `\class@info`

A few shorthands for Class messages. Your document class should define `\class@name`.

```
\def\class@err#1{\ClassError{\class@name}{#1}\@eha}%
\def\class@warn#1{\ClassWarningNoLine{\class@name}{#1}}%
\def\class@info#1{\ClassInfo{\class@name}{#1}}%
\def\obsolete@command#1{%
 \class@warn@end{Command \string#1\space is obsolete.^^JPlease remove from your document}%
 \global\let#1\@empty
 #1%
}%
\def\replace@command#1#2{%
 \class@warn@end{Command \string#1\space is obsolete;^^JUse \string#2\space instead}%
 \global\let#1#2%
 #1%
}%
\def\replace@environment#1#2{%
```

```
 \class@warn@end{Environment #1 is obsolete;^^JUse #2 instead}%
 \glet@environment{#1}{#2}%
 \@nameuse{#1}%
}%
\def\incompatible@package#1{%
 \@ifpackageloaded{#1}{%
  \def\@tempa{I cannot continue. You must remove the \string\usepackage\ statement that caused
  \ClassError{\class@name}{The #1 package cannot be used with \class@name}%
  \@tempa\stop
 }{%
  \class@info{#1 was not loaded (OK!)}%
 }%
}%
\def\class@warn@end#1{%
 \gappdef\class@enddocumenthook{\class@warn{#1}}%
}%
\AtEndOfClass{%
 \@ifxundefined\class@name{\def\class@name{Generic Class}}{}%
}%
```

6.4 New Tools

`\t@`

```
\def\t@{to}%
```

`\dimen@iii`

```
\dimendef\dimen@iii\thr@@
```

`\halignt@`

```
\def\halignt@{\halign\t@}%
```

`\f@ur` Analogous to `\@ne`, `\tw@`, and `\thr@@`.

```
\chardef\f@ur=4\relax
\chardef\cat@letter=11\relax
\chardef\other=12\relax
```

`\let@environment` `\glet@environment` The directive `\let@environment` takes care of a common programming idiom whereby one environment is made a synonym for another.

```
\def\let@environment#1#2{%
 \expandafter\let
 \csname#1\expandafter\endcsname\csname#2\endcsname
 \expandafter\let
 \csname end#1\expandafter\endcsname\csname end#2\endcsname
}%
\def\glet@environment#1#2{%
 \global\expandafter\let
 \csname#1\expandafter\endcsname\csname#2\endcsname
 \global\expandafter\let
 \csname end#1\expandafter\endcsname\csname end#2\endcsname
}%
```

\tracingplain The command \tracingplain causes TEX's tracing parameters to return to the values set by default. This command is sometimes useful when you have said \tracingall somewhere and want to restore. The \traceoutput command causes \tracingoutput diagnostics upon \shipout.

```
\newcommand\tracingplain{%
 \tracingonline\z@\tracingcommands\z@\tracingstats\z@
 \tracingpages\z@\tracingoutput\z@\tracinglostchars\@ne
 \tracingmacros\z@\tracingparagraphs\z@\tracingrestores\z@
 \showboxbreadth5\showboxdepth3\relax %\errorstopmode
 }%
\newcommand\traceoutput{%
 \appdef\@resetactivechars{\showoutput}%
}%
```

\say \saythe The commands \say and \saythe cause diagnostic messages in the TEX log that give the value of a control sequence name or a register respectively.

```
\newcommand\say[1]{\typeout{<\noexpand#1=\meaning#1>}}%
\newcommand\saythe[1]{\typeout{<\noexpand#1=\the#1>}}%
```

\fullinterlineskip Resets the \prevdepth so that the full amount of \baselineskip glue will be inserted by the \baselinesklip mechanism. Can be invoked just after a \hrule to undo its default suppression of base line skip.

```
\def\fullinterlineskip{\prevdepth\z@}%
```

\count@i \count@ii

```
\countdef\count@i\@ne
\countdef\count@ii\tw@
```

6.5 Boolean Control

We introduce just enough of the Boolean calculus for TEX. Alan Jeffrey was the pioneer here, with an article in TUGboat (Vol. 11, No. 2, page 237). This implementation owes a debt to William Baxter (web@superscript.com). See articles by Baxter and Ogawa in the proceedings of the 1994 TUG meeting, TUGboat Vol. 15, No. 3.

\prepdef \appdef \gappdef Provide the capability of performing head- and tail patches. The procedure \prepdef prepends to the given macro the tokens specified in its second argument. Likewise for \appdef, except that it appends. Note that the first 10 toks registers are utility registers, and we simply make a control sequence name, \toks@ii, for one of them.

```
\long\def\prepdef#1#2{%
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \edef#1{\the\toks@ii\the\toks@}%
}%
\long\def\appdef#1#2{%
```

```
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \edef#1{\the\toks@\the\toks@ii}%
}%
\long\def\gappdef#1#2{%
 \@ifxundefined#1{\toks@{}}{\toks@\expandafter{#1}}%
 \toks@ii{#2}%
 \global\edef#1{\the\toks@\the\toks@ii}%
}%
\long\def\appdef@val#1#2{%
 \appdef#1{{#2}}%
}%
\long\def\appdef@e#1#2{%
 \expandafter\appdef
 \expandafter#1%
 \expandafter{#2}%
}%
\long\def\appdef@eval#1#2{%
 \expandafter\appdef@val
 \expandafter#1%
 \expandafter{#2}%
}%
\toksdef\toks@ii=\tw@
```

`\@ifxundefined` `\@ifnotrelax` `\@argswap` `\@argswap@val`

Certain utility procedures use `\@ifxundefined`, which is defined here in terms of `\@ifx`. Others use `\@ifnotrelax`, namely when the control sequence name is manufactured by the use of `\csname`.

The procedures `\@argswap`and `\@argswap@val`are used to facilitate control of expansion.

```
\long\def\@ifxundefined#1{\@ifx{\undefined#1}}%
\long\def\@ifnotrelax#1#2#3{\@ifx{\relax#1}{#3}{#2}}%
\long\def\@argswap#1#2{#2#1}%
\long\def\@argswap@val#1#2{#2{#1}}%
\def\@ifxundefined@cs#1{\expandafter\@ifx\expandafter{\csname#1\endcsname\relax}}%
```

`\rvtx@ifformat@geq`

Some changes in the LaTeX kernel requires us to conditionally define some macros depending on the version of the kernel. `\rvtx@ifformat@geq` will check if the release date of the currently-running LaTeX 2$_\varepsilon$ kernel is greater or equal to the argument (the argument should be in the format `yyyy-mm-dd`).

```
\ifx\IfFormatAtLeastTF\undefined
  \def\rvtx@ifformat@geq{\@ifl@t@r\fmtversion}%
\else
  \let\rvtx@ifformat@geq\IfFormatAtLeastTF
\fi
```

`\@boolean` `\@boole@def`

In order to define `\@ifx`, we first must create the "defining word" (term taken form our Forth vocabulary) `\@boole@def`, which employs `\@boolean` to do its job.

```
\def\@boolean#1#2{%
  \long\def#1{%
```

```
    #2% \if<something>
      \expandafter\true@sw
    \else
      \expandafter\false@sw
    \fi
  }%
}%
\def\@boole@def#1#{\@boolean{#1}}% Implicit #2
```

`\@booleantrue` `\@booleanfalse` The procedures `\@booleantrue` and `\@booleanfalse` are assignment operators for Boolean flags.

```
\def\@booleantrue#1{\let#1\true@sw}%
\def\@booleanfalse#1{\let#1\false@sw}%
```

`\@ifx` `\@ifx@empty` `\@if@empty` `\@ifcat` `\@ifdim` `\@ifeof` `\@ifhbox` `\@ifhmode` `\@ifinner` `\@ifmmode` `\@ifnum` `\@ifodd` `\@ifvbox` `\@ifvmode` `\@ifvoid` We can now invoke the defining word to create the procedures `\@ifx` and friends.

Compatability Note: earlier versions of this package defined a procedure `\@ifempty`. However, for compatability with AMSLaTeX, we must avoid the following three names: `\@ifempty`, `\@xifempty`, and `\@ifnotempty`.

```
\@boole@def\@ifx#1{\ifx#1}%
\@boole@def\@ifx@empty#1{\ifx\@empty#1}%
\@boole@def\@if@empty#1{\if!#1!}%
%\@boole@def\@if@sw#1{\csname if#1\endcsname}%
\def\@if@sw#1#2{#1\expandafter\true@sw\else\expandafter\false@sw#2}%
\@boole@def\@ifdim#1{\ifdim#1}%
\@boole@def\@ifeof#1{\ifeof#1}%
\@boole@def\@ifhbox#1{\ifhbox#1}%
\@boole@def\@ifhmode{\ifhmode}%
\@boole@def\@ifinner{\ifinner}%
\@boole@def\@ifmmode{\ifmmode}%
\@boole@def\@ifnum#1{\ifnum#1}%
\@boole@def\@ifodd#1{\ifodd#1}%
\@boole@def\@ifvbox#1{\ifvbox#1}%
\@boole@def\@ifvmode{\ifvmode}%
\@boole@def\@ifvoid#1{\ifvoid#1}%
```

`\true@sw` `\false@sw` Note that when a Boolean operator expands, it employs two macros that act as selectors, defined here.

```
\long\def\true@sw#1#2{#1}%
\long\def\false@sw#1#2{#2}%
```

`\loopuntil` `\loopwhile` Loop control using the Boolean idiom. Superior to `\loop`...`\repeat` because these can be nested. The tail of the argument must have a Boolean predicate.

```
\long\def\loopuntil#1{#1{}{\loopuntil{#1}}}%
\long\def\loopwhile#1{#1{\loopwhile{#1}}{}}%
```

`\@provide` A defining word that refuses to clobber a prior meaning.

```
\def\@provide#1{%
 \@ifx{\undefined#1}{\true@sw}{\@ifx{\relax#1}{\true@sw}{\false@sw}}%
 {\def#1}{\def\j@nk}%
}%
```

6.6 Begin Document Structure

The standard LaTeX mechanism `\AtBeginDocument` is inadequate because the `\vsize` is bound much too early. We supply here a mechanism whereby decisions about the page layout can be deferred until `\AtBeginDocument` time.

The problem we are working around is that the `\AtBeginDocument` hook in `\document` appears long after the calculation of `\vsize` and `\hsize`, that is, LaTeX provides no mechanism for deferring the decision about the page grid until `\AtBeginDocument` time. We fix things by prepending a hook at the very beginning of `\document`.

`\document` We begin by installing hooks into `\document` that we will manage ourselves.

The 2020-10-01 LaTeX release got a new hook management system and several new hooks (several previously provided by etoolbox). The one we want here is `begindocument/before`, the first thing executed by `\document`, right after ending the group started by `\begin`.

Thus, if the LaTeX kernel date is 2020-10-01 we just add to that hook, otherwise resort to the old method, patching `\document`: end the group started by `\begin`, apply our hook, and conclude our shenanigans by absorbing the first token of the expansion of `\document`, which we assume to be `\endgroup` (true until the aforementioned release).

```
\rvtx@ifformat@geq{2020-10-01}%
  {%
    \AddToHook{begindocument/before}{%
      \init@documenthook
      \set@typesize@hook
      \normalsize
      \set@pica@hook
    }%
  }{%
    \prepdef\document{%
     \endgroup
     \init@documenthook
     \set@typesize@hook
     \normalsize
     \set@pica@hook
     \true@sw{}%
    }%
  }
```

`\class@documenthook` We install the first `\AtBeginDocument` hook, namely the procedure `\class@documenthook`.
`\class@enddocumenthook` Within the document class, we will use this hook exclusively, so as to avoid interference from other packages. Similarly with `\class@enddocumenthook`, installed via `\AtEndDocument`.

A document class using this package should do as this package does and just say, `\appdef\class@documenthook` and `\appdef\class@enddocumenthook` instead of `\AtBeginDocument` and `\AtEndDocument`.

```
\def\init@documenthook{}%
```

```
\AtBeginDocument{%
 \class@documenthook
}%
\AtEndDocument{%
 \class@enddocumenthook
}%
\def\class@documenthook{}%
\def\class@enddocumenthook{}%
```

`\set@typesize@hook` `\set@pica@hook`

The macros `\set@typesize@hook` and `\set@pica@hook` provide everything we need. To use, simply `\appdef` your tokens to the appropriate hook.

```
\def\set@typesize@hook{}%
\def\set@pica@hook{}%
```

`\enddocument` `\check@aux` `\do@check@aux`

The standard LaTeX `\end{document}` processing is a potential problem, particularly when the output routine has been changed by `ltxgrid`. We separate out the procedure that checks the auxiliary file at the end of the job so that later it can be called from the safety of the output routine. We will do this to ensure that the `\@mainaux` stream is not closed until the last page of the job is shipped out, and that can only be done by coordinating with the output routine.

This approach, however, will only be done for older versions of the LaTeX kernel:

```
\rvtx@ifformat@geq{2020-10-01}{%
  % <definitions for newer LaTeX later>
}{%
  % <definitions for older LaTeX>
\def\enddocument{%
 \@enddocumenthook
 \@checkend{document}%
 \clear@document
 \check@aux
 \deadcycles\z@
 \@@end
}%
\def\clear@document{\clearpage}%
\def\check@aux{\do@check@aux}%
\def\do@check@aux{%
 \@if@sw\if@filesw\fi{%
  \immediate\closeout\@mainaux
  \let\@setckpt\@gobbletwo
  \let\@newl@bel\@testdef
  \@tempswafalse
  \makeatletter
  \input\jobname.aux\relax
 }{}%
 \@dofilelist
 \@ifdim{\font@submax >\fontsubfuzz\relax}{%
  \@font@warning{%
   Size substitutions with differences\MessageBreak
   up to \font@submax\space have occured.\@gobbletwo
```

```
  }%
 }{}%
 \@defaultsubs
 \@refundefined
 \@if@sw\if@filesw\fi{%
  \@ifx{\@multiplelabels\relax}{%
   \@if@sw\if@tempswa\fi{%
    \@latex@warning@no@line{%
     Label(s) may have changed.
     Rerun to get cross-references right%
    }%
   }{}%
  }{%
    \@multiplelabels
  }%
 }{}%
}%
}
```

`\rvtx@enddocument@patch` For newer LaTeX we'll try to be a bit more future-proof (no miracle though). The code for `\enddocument` (in pre-2020-10-01 LaTeX) is roughly:

```
%   \def\enddocument{%
%     ⟨hooks and bookkeeping⟩
%     \clearpage
%     ⟨read main .aux and final checks⟩
%     \@@end
%   }
%
```

and the patches above replace the `\clearpage` by its own `\clear@document`, and `<read main .aux and final checks>` by `\do@check@aux`, which it can later control the timing.

Now we will apply the same changes, but this time without redefining `\enddocument`: we will instead replace tokens on-the-fly, when `\enddocument` is expanded. This will grant us a slightly safer approach that won't depend so much on the internals of `\enddocument`.

This entire patch should work with the previous definition of `\enddocument` as well (except it cannot be used in the hook), but for now leave previous versions untouched.

The entire patching will reside in the `enddocument` hook:

```
\rvtx@ifformat@geq{2020-10-01}{%
 \AddToHook{enddocument}{\rvtx@enddocument@patch{}}%
}{}
```

This macro will be executed after `\enddocument` has expanded, so all its tokens are now exposed. Here we will assume that `\enddocument` contains the tokens `\@checkend{document}` and `\endgroup`, and use them as delimiters:

```
\protected\long\def\rvtx@enddocument@patch#1#2\@checkend#3{%
```

```
  \begingroup
    \edef\x{\detokenize{#3}}%
    \edef\y{\detokenize{document}}%
  \expandafter\endgroup
  \ifx\x\y
    \expandafter\rvtx@enddocument@patch@end
  \else
    \expandafter\rvtx@enddocument@patch@more
  \fi
    {#1#2}{#3}}
\def\rvtx@enddocument@patch@more#1#2{%
  \rvtx@enddocument@patch{#1\@checkend{#2}}}
```

When the `\@checkend{document}` is reached, use `\clearpage` and `\enddocument` as delimiters for the `<read main .aux and final checks>` part, and save it in `\do@check@aux`:

```
\long\def\rvtx@enddocument@patch@end#1#2\clearpage#3\endgroup{%
  \def\do@check@aux{#3\endgroup}%
```

Then execute the code consumed in the previous step:

```
  #1%
  \@checkend{#2}%
```

Do `\clear@document` instead of `\clearpage` and `\check@aux` instead of the code grabbed.

```
  \clear@document
  \check@aux}
\def\check@aux{\do@check@aux}%
\def\clear@document{\clearpage}%
```

6.7 Type Tools

`\flushing` Undoes `\centering`. Should also undo `\raggedleft` and `\raggedright`.

```
\def\flushing{%
  \let\\\@normalcr
  \leftskip\z@skip
  \rightskip\z@skip
  \@rightskip\z@skip
  \parfillskip\@flushglue
}%
```

6.8 Display Math

`\eqnarray@LaTeX` `\eqnarray@fleqn@fixed` Team LaTeX has stated they will never repair Leslie's broken definition of `eqnarray`. Let us be bold....

Note on `hyperref` package compatability: that package overrides `\eqnarray` by wrapping it up in a larger procedure, so its changes are compatible with this package's changes.

```
\def\eqnarray@LaTeX{%
```

```
   \stepcounter{equation}%
   \def\@currentlabel{\p@equation\theequation}%
   \global\@eqnswtrue
   \m@th
   \global\@eqcnt\z@
   \tabskip\@centering
   \let\\\@eqncr
   $$\everycr{}\halign\t@\displaywidth\bgroup
       \hskip\@centering$\displaystyle\tabskip\z@skip{##}$\@eqnsel
      &\global\@eqcnt\@ne\hskip \tw@\arraycolsep \hfil${##}$\hfil
      &\global\@eqcnt\tw@ \hskip \tw@\arraycolsep
         $\displaystyle{##}$\hfil\tabskip\@centering
      &\global\@eqcnt\thr@@ \hb@xt@\z@\bgroup\hss##\egroup
         \tabskip\z@skip
      \cr
}
\long\def\eqnarray@fleqn@fixed{%
 \stepcounter{equation}\def\@currentlabel{\p@equation\theequation}%
 \global\@eqnswtrue\m@th\global\@eqcnt\z@
 \tabskip\mathindent
 \let\\=\@eqncr
 \setlength\abovedisplayskip{\topsep}%
 \ifvmode\addtolength\abovedisplayskip{\partopsep}\fi
 \addtolength\abovedisplayskip{\parskip}%
 \setlength\belowdisplayskip{\abovedisplayskip}%
 \setlength\belowdisplayshortskip{\abovedisplayskip}%
 \setlength\abovedisplayshortskip{\abovedisplayskip}%
 $$%
 \everycr{}%
 \halignt@\linewidth\bgroup
  \hskip\@centering$\displaystyle\tabskip\z@skip{##}$\@eqnsel
  &\global\@eqcnt\@ne
   \hskip\tw@\eqncolsep
   \hfil${{}##{}}$\hfil
  &\global\@eqcnt\tw@
   \hskip\tw@\eqncolsep
   $\displaystyle{##}$\hfil\tabskip\@centering
  &\global\@eqcnt\thr@@\hb@xt@\z@\bgroup\hss##\egroup
   \tabskip\z@skip
  \cr
}%
\@ifx{\eqnarray\eqnarray@LaTeX}{%
 \class@info{Repairing broken LaTeX eqnarray}%
 \let\eqnarray\eqnarray@fleqn@fixed
 \newlength\eqncolsep
 \setlength\eqncolsep\z@
 \let\eqnarray@LaTeX\relax
 \let\eqnarray@fleqn@fixed\relax
}{}%
\def\mathindent{\@centering}%
```

```
\def\set@eqnarray@skips{}%
```

6.9 Footnotes

\footnote
\footnotemark
\@xfootnote
\@xfootnotemark
\@yfootnote

We repair an error in the LaTeX kernel (see `ltfloat.dtx`) involving footnotes. The symptom is that the `\footnotemark` command does not work properly within a `minipage` environment. The source of the problem is in the way the `\footnotemark` and `\@xfootnotemark` procedures are defined: they do not share the method used by the `\footnote` and other procedures that allows a context switch to change the way footnotes behave within a minipage environment. This is a LaTeX bug of long standing; this fix dates to 1987.

While we are at it, we rewrite both the `\footnote` and `\footnotemark` procedures, achieving a slightly cleaner separation of syntax and semantics. Note that the `\@footnotemark` and `\@footnotetext` procedures are not altered here; they continue as the methods of formatting the footnote mark and footnote text, respectively.

A note about the context switch mentioned above: the `minipage` environment executes the following in order to alter the way footnotes behave:

```
%\def\@mpfn{mpfootnote}%
%\def\thempfn{\thempfootnote}%
%\let\@footnotetext\@mpfootnotetext
%\c@mpfootnote\z@
%
```

This code changes the counter used in autonumbered footnotes, the choice of footnote marker, and the procedure used on the footnote text. Changing the counter is needed because minipage footnotes are in their own sequence, and the footnote marker is customarily different within a minipage. The procedure that works on the footnote text must be different because the footnotes are placed at the bottom of the minipage, not the bottom of the text column.

Any procedure that establishes a minipage-like context (e.g., floats) can do the same.

```
\def\footnote{%
 \@ifnextchar[\@xfootnote{\@yfootnote\@footnotetext}%
}%
\def\footnotemark{%
 \@ifnextchar[\@xfootnotemark{\@yfootnote}%
}%
\def\@xfootnote[#1]{%
 \@xfootnotemark[#1]%
 \@footnotetext
}%
\def\@xfootnotemark@ltx[#1]{%
 \begingroup
   \csname c@\@mpfn\endcsname #1\relax
   \unrestored@protected@xdef\@thefnmark{\thempfn}%
 \endgroup
```

```
 \H@@footnotemark
}%
\def\@yfootnote{%
 \stepcounter\@mpfn
 \protected@xdef\@thefnmark{\thempfn}%
 \H@@footnotemark
}%
```

Note on `hyperref` compatability: In its "Automated LaTeX hypertext cross-references", the `hyperref` package alters footnote processing, thereby imperiling these fixes and necessiating defensive measures.

The main thing `hyperref` does is to take over the `\@mpfootnotetext` and `\@footnotetext` procedures, feeding its own arguments to these macros. It also rewrites `\@footnotemark`, making it a hyperlink.

But at the same time, it attempts to turn off these changes during `\maketitle` processing, necessitating rewriting `\@xfootnotemark`. At this point it is on the slippery slope.

We make ourself `hyperref` friendly: we give `hyperref` what it needs, but avoid its change to `\@xfootnotemark`.

Any other package that rewrites LaTeX's footnote macros will be incompatible with this package.

```
\appdef\class@documenthook{%
 \@ifxundefined\H@@footnotemark{%
  \let\H@@footnotemark\@footnotemark
 }{}%
 \let\@xfootnotemark\@xfootnotemark@ltx
}%
```

Two thoughts about `hyperref`: what for does it define `\realfootnote`? Also: a document class that desires high hypertext capabilities might well wish to reimplement `\maketitle` so that footnotes called out from there are hypertext links: the `hyperref` package's "Automated LaTeX hypertext cross-references" does not do any of this:

> But the special footnotes in `\maketitle` are much too hard to deal with properly. Let them revert to plain behaviour.

Note that the document class, in reimplementing `\maketitle`, must ensure that the `hyperref` package does not clobber its own definition!

`\@footnotetext`
`\@mpfootnotetext`
`\@tpfootnotetext`
`\make@footnotetext`
`\set@footnotewidth`

The two procedures `\@footnotetext` and `\@mpfootnotetext` share code. We make that explicit here.

Note that the procedure calling `\make@footnotetext` will open a group with `\bgroup` which is then closed by `\minipagefootnote@drop`.

Difference from LaTeX: here we do not set `\floatingpenalty` to infinity. Doing this must date back to a time when LaTeX could not accomodate split insertions (footnotes). I cannot think of any other reason to do have done this. At any rate, with the `ltxgrid` package, split insertions are specifically properly taken care of, so we allow it.

We provide the hook `\set@footnotewidth` that sets the footnote on a particular measure. Some page grids are such as to set a footnote in a context where `\columnwidth`is not the right parameter to use for the set width of a footnote. In such a case, for the applicable scope, you should define `\set@footnotewidth` to perform this job correctly.

A procedure, `\set@footnotewidth@ii`, illustrates how to do this when in a two-column page grid. In general, remember that footnotes, like all insertions (including floats), are a step outside of the galley context, and all aspects of insertions need to be properly handled, including the set width.

```
\long\def\@footnotetext{%
 \insert\footins\bgroup
  \make@footnotetext
}%
\long\def\@mpfootnotetext{%
 \minipagefootnote@pick
  \make@footnotetext
}%
\def\make@footnotetext#1{%
  \reset@font\footnotesize
  \interlinepenalty\interfootnotelinepenalty
  \splittopskip\footnotesep
  \splitmaxdepth\dp\strutbox
% \floatingpenalty\@MM
  \set@footnotewidth
  \@parboxrestore
  \protected@edef\@currentlabel{%
   \csname p@footnote\endcsname\@thefnmark
  }%
  \color@begingroup
   \@makefntext{%
    \rule\z@\footnotesep\ignorespaces#1\@finalstrut\strutbox
   }%
  \color@endgroup
 \minipagefootnote@drop
}%
\def\set@footnotewidth{%
 \hsize\columnwidth
 \linewidth\hsize
}%
\def\set@footnotewidth@ii{%
 \hsize\textwidth
 \advance\hsize\columnsep
 \divide\hsize\tw@
 \advance\hsize-\columnsep
 \linewidth\hsize
}%
```

6.10 Floats

6.10.1 Usage notes

We extend the LaTeX kernel for three purposes:

1. When the `\footnote` command is used within the scope of a float, we do as `minipage` does.

2. We provide a mechanism to write floats out to an external stream for temporary storage (deferred floats).

3. We provide mechanism for placing a float `here` invariably, that is, floats are unfloated. This mechanism is used to read the external stream mentioned above.

To use these mechanisms, the document class should define a float, say, `figure` as per usual, and in addition:

1. Optionally define an alternative, say `figure@write` as follows:

   ```
   %\newenvironment{figure@write}{%
   % \write@float{figure}%
   %}{%
   % \endwrite@float
   %}%
   %
   ```

   That is, the alternative environment executes `\write@float` instead of `\@float`. Note that this step is not needed if the float environment is defined in the simple way of `classes.dtx`. However, an environment like `longtable` will require it.

2. Install into `\AtBeginDocument` a call to `\do@if@floats`, with the float name and an appropriate file extension as its arguments.

   ```
   %\AtBeginDocument{\do@if@floats{figure}{.fgx}}%
   %
   ```

3. Optionally define a text entity `\figuresname` that will be the text of the head that is set over the deferred floats. If not defined, there will be no head.

4. Optionally define a user-level command to allow the document to determine where the figures are printed out (default is to print at end of document). E.g.,

   ```
   %\newcommand\printfigures{\print@float{figure}}%
   %
   ```

5. Install into `\appdef\class@enddocumenthook` a call to `\printfigures`, or, if the latter is not defined, as follows:

```
%\appdef\class@enddocumenthook{\print@float{figure}}%
%
```

Note that installing this command into `\AtBeginDocument`is best done earlier than calls that assume the last page of the document is at hand.

6.10.2 Robustifying fragile commands

Certain of LaTeX's commands cannot be written out to a file or appear within a `\mark` command argument because they do calculations during expansion. We provide for a little help, but without changing the meanings of these commands.

`\addtocontents`

```
\def\addtocontents#1#2{%
 \protected@write\@auxout{%
  \let \label \@gobble \let \index \@gobble \let \glossary \@gobble
  \def\({\string\(}%
  \def\){\string\)}%
  \def\\{\string\\}%
 }{\string \@writefile {#1}{#2}}%
}%
```

6.10.3 Preparing for the `hyperref` package

`\addcontentsline` `\contentsline`

The `hyperref` package assumes that the `\contentsline` command will be given four arguments. Therefore it cannot successfully process a ltxutil.dtx.toc file that had been written by standard LaTeX. We fix things up by always writing that fourth argument and by supplying a `\contentsline` command that can read them.

We also give the `\newlabel` command's second argument five tokens.

This means that a document class that uses this package will itself have trouble taking over a ltxutil.dtx.toc file that was written by standard LaTeX. Sigh.

```
\def\addcontentsline#1#2#3{%
 \addtocontents{#1}{%
  \protect\contentsline{#2}{#3}{\thepage}{}%
 }%
}%
\def\contentsline#1#2#3#4{%
 \csname l@#1\endcsname{#2}{#3}%
}%
\def\label#1{%
 \@bsphack
  \protected@write\@auxout{}{%
   \string\newlabel{#1}{{\@currentlabel}{\thepage}{}{}{}}%
  }%
 \@esphack
}%
```

6.10.4 Footnotes within floats, unfloating floats, float font

\caption DPC: Er a bit of a hack, but seems best way of supporting normal LaTeX syntax at this point: If a caption is used below a table, then put out the footnotes before the caption.

```
\appdef\class@documenthook{%
 \prepdef\caption{\minipagefootnote@here}%
}%
```

Note on `hyperref` compatability: this change to the `\caption` command is compatible with the "Automated LaTeX hypertext cross-references" patches of that package.

All the same, I think Sebastian's changes to `\caption` and `\@caption` could bear with some improvement. The following implementation requires knowing only the pattern part of the `\@caption` macro:

```
%\def\caption{%
%  \H@refstepcounter\@captype
%  \hyper@makecurrent{\@captype}%
%  \@dblarg{\H@caption\@captype}%
%}%
%\def\H@caption#1[#2]#3{%
% \@caption{#1}[#2]{%
%  \ifHy@nesting
%   \hyper@@anchor{\@currentHref}{#3}%
%  \else
%   \hyper@@anchor{\@currentHref}{\relax}#3%
%  \fi
% }%
%}%
%
%
```

\minipagefootnote@init
\minipagefootnote@here
\minipagefootnote@foot
\minipagefootnote@pick
\minipagefootnote@drop

Procedure to deal with footnotes accumulated within a minipage environment. These procedures encapsulate all uses of the `\@mpfootins` box.

Note: `\minipagefootnote@here` must *not* be executed within the MVL!

```
\def\minipagefootnote@init{%
 \setbox\@mpfootins\box\voidb@x
}%
\def\minipagefootnote@pick{%
 \global\setbox\@mpfootins\vbox\bgroup
  \unvbox\@mpfootins
}%
\def\minipagefootnote@drop{%
 \egroup
}%
\def\minipagefootnote@here{%
    \par
    \@ifvoid\@mpfootins{}{%
      \vskip\skip\@mpfootins
```

```
      \fullinterlineskip
      \@ifinner{%
       \vtop{\unvcopy\@mpfootins}%
       {\setbox\z@\lastbox}%
      }{}%
      \unvbox\@mpfootins
    }%
}%
\def\minipagefootnote@foot{%
 \@ifvoid\@mpfootins{}{%
  \insert\footins\bgroup\unvbox\@mpfootins\egroup
 }%
}%
\def\endminipage{%
    \par
    \unskip
    \minipagefootnote@here
    \@minipagefalse   %% added 24 May 89
  \color@endgroup
  \egroup
  \expandafter\@iiiparbox\@mpargs{\unvbox\@tempboxa}%
}%
```

`\floats@sw` The Boolean `\floats@sw` signifies that floats are to be floated; if false, that floats are to be deferred to the end of the document. Note that the state of this Boolean is to be changed by the document class in response to user-selected options. Here we display model code that assigns a default value at `\AtBeginDocument` time.

```
%\AtBeginDocument{%
% \@ifxundefined\floats@sw{\@booleantrue\floats@sw}{}%
%}%
%
```

`\@xfloat` `\@mpmakefntext` The float start-code is redefined to set up footnotes in the style of minipage. Also, the `\floats@sw` Boolean informs us that floats are to be all placed `here`. Note that, to protect against the Boolean being undefined at this late hour, we default it globally to true.

FIXME: why does hyperref override `\@xfootnotenext`?

```
\let\@xfloat@LaTeX\@xfloat
\def\@xfloat#1[#2]{%
  \@xfloat@prep
  \@nameuse{fp@proc@#2}%
  \@ifxundefined\floats@sw{\global\@booleantrue\floats@sw}{}%
  \floats@sw{\@xfloat@LaTeX{#1}[#2]}{\@xfloat@anchored{#1}[]}%
}%
\def\@xfloat@prep{%
  \let\footnote\footnote@latex
  \def\@mpfn{mpfootnote}%
  \def\thempfn{\thempfootnote}%
```

```
% \def\thefootnote{\thempfootnote}%
  \c@mpfootnote\z@
  \let\@footnotetext\@mpfootnotetext
  \let\H@@footnotetext\@mpfootnotetext
  \let\@makefntext\@mpmakefntext
% \samepage
}%
\appdef\class@documenthook{%
 \let\footnote@latex\footnote
}%
%\def\fp@proc@h{\@booleanfalse\floats@sw}%
%\def\fp@proc@H{\@booleanfalse\floats@sw}%
\def\@xfloat@anchored#1[#2]{%
 \def\@captype{#1}%
 \begin@float@pagebreak
%\vbox\bgroup
  \let\end@float\end@float@anchored
  \let\end@dblfloat\end@float@anchored
% do as \@xfloat does:
        \hsize\columnwidth
        \@parboxrestore
        \@floatboxreset
  \minipagefootnote@init
% \pagegrid@col\@ne % Klootch to avoid processing as a float
}%
\def\end@float@anchored{%
  \minipagefootnote@here
  \par\vskip\z@skip      %% \par\vskip\z@ added 15 Dec 87
%\egroup
 \par
 \end@float@pagebreak
}%
\def\begin@float@pagebreak{\par\addvspace\intextsep}%
\def\end@float@pagebreak{\par\addvspace\intextsep}%
\def\@mpmakefntext#1{%
 \parindent=1em
 \noindent
 \hb@xt@1em{\hss\@makefnmark}%
 #1%
}%
```

6.10.5 Writing floats out to a file

`\do@if@floats` The procedure `\do@if@floats` should be executed at `\AtBeginDocument` time, and arranges to write out the floats of the given class to a temporary file, to be read back later (deferred floats), given that `\floats@sw` is false. Note that, to protect against the Boolean being undefined at this late hour, we default it globally to true.

```
\def\do@if@floats#1#2{%
```

```
\@ifxundefined\floats@sw{\global\@booleantrue\floats@sw}{}%
\floats@sw{}{%
```

Open the stream to save out the document's floats of this class.

```
 \expandafter\newwrite
             \csname#1write\endcsname
 \expandafter\def
             \csname#1@stream\endcsname{\jobname#2}%
 \expandafter\immediate
 \expandafter\openout
             \csname#1write\endcsname
             \csname#1@stream\endcsname\relax
```

Swap environments. If the class writer has defined, e.g., `figure@write`, then we use this as the procedure to execute for writing the float out to the external stream. Otherwise, the replacement of `\@float` by `\write@float` should do the right thing for float environments defined in the simple way of `classes.dtx`.

```
 \@ifxundefined\@float@LaTeX{%
  \let\@float@LaTeX\@float
  \let\@dblfloat@LaTeX\@dblfloat
  \let\@float\write@float
  \let\@dblfloat\write@floats
 }{}%
 \let@environment{#1@float}{#1}%
 \let@environment{#1@floats}{#1*}%
 \@ifxundefined@cs{#1@write}{}{%
  \let@environment{#1}{#1@write}%
 }%
}%
```

`\print@float` The procedure `\print@float` prints out the deferred floats.

```
\def\triggerpar{\leavevmode\@@par}%
\def\oneapage{\def\begin@float@pagebreak{\newpage}\def\end@float@pagebreak{\newpage}}%
\def\print@float#1#2{%
 \@ifxundefined@cs{#1write}{}{%
  \begingroup
   \@booleanfalse\floats@sw
   #2%
   \raggedbottom
   \def\array@default{v}% floats must
   \let\@float\@float@LaTeX
   \let\@dblfloat\@dblfloat@LaTeX
   \let\trigger@float@par\triggerpar
   \let@environment{#1}{#1@float}%
   \let@environment{#1*}{#1@floats}%
   \expandafter\prepdef\csname#1\endcsname{\trigger@float@par}%
   \expandafter\prepdef\csname#1*\endcsname{\trigger@float@par}%
   \@namedef{fps@#1}{h!}%
   \expandafter\immediate
```

```
   \expandafter\closeout
               \csname#1write\endcsname
   \everypar{%
    \global\let\trigger@float@par\relax
    \global\everypar{}\setbox\z@\lastbox
    \@ifxundefined@cs{#1sname}{}{%
     \begin@float@pagebreak
     \expandafter\section
     \expandafter*%
     \expandafter{%
                  \csname#1sname\endcsname
                 }%
    }%
   }%
   \input{\csname#1@stream\endcsname}%
  \endgroup
  \global\expandafter\let\csname#1write\endcsname\relax
 }%
}%
```

`\write@float` `\write@floats` `\write@@float`

Handles the case where the name of the float is the same as that of the stream. Note that `longtable` does *not* fit this case. Note also: `\write@float` is *not* a user-level environment therefore it is properly not defined with `\newenvironment`.

```
\def\write@float#1{\write@@float{#1}{#1}}%
\def\endwrite@float{\@Esphack}%
\def\write@floats#1{\write@@float{#1*}{#1}}%
\def\endwrite@floats{\@Esphack}%
```

`\write@@float`

```
\def\write@@float#1#2{%
  \ifhmode
     \@bsphack
  \fi
  \chardef\@tempc\csname#2write\endcsname
  \toks@{\begin{#1}}%
  \def\@tempb{#1}%
  \expandafter\let\csname end#1\endcsname\endwrite@float
  \catcode`\^^M\active
  \@makeother\{\@makeother\}\@makeother\%
  \write@floatline
}%
```

`\write@floatline` `\@write@floatline` `\float@end@tag`

The procedure `\write@floatline` only parses, and passes its result to `\@write@floatline`, which writes the line to output, then tests the line for the `\end{`⟨*float*⟩`}` tokens with aid of the `\float@end@tag` procedure.

```
\begingroup
 \catcode`\[\the\catcode`\{\catcode`\]\the\catcode`\}\@makeother\{\@makeother\}%
 \gdef\float@end@tag#1\end{#2}#3\@nul[%
  \def\@tempa[#2]%
```

```
  \@ifx[\@tempa\@tempb][\end[#2]][\write@floatline]%
 ]%
 \obeylines%
 \gdef\write@floatline#1^^M[%
  \begingroup%
   \newlinechar`\^^M%
   \toks@\expandafter[\the\toks@#1]\immediate\write\@tempc[\the\toks@]%
  \endgroup%
  \toks@[]%
  \float@end@tag#1\end{}\@nul%
 ]%
\endgroup
```

6.11 Counters

The following definitions override those of the LaTeX kernel, providing for a greater range of inputs.

```
\def\@alph#1{\ifcase#1\or a\or b\or c\or d\else\@ialph{#1}\fi}
```

```
\def\@ialph#1{\ifcase#1\or \or \or \or \or e\or f\or g\or h\or i\or j\or
  k\or l\or m\or n\or o\or p\or q\or r\or s\or t\or u\or v\or w\or x\or
  y\or z\or aa\or bb\or cc\or dd\or ee\or ff\or gg\or hh\or ii\or jj\or
  kk\or ll\or mm\or nn\or oo\or pp\or qq\or rr\or ss\or tt\or uu\or
  vv\or ww\or xx\or yy\or zz\else\@ctrerr\fi}
```

6.12 Customization of Sections

Patch the standard LaTeX sectioning procedure to:

- Allow a sectioning command to trigger the title page, or more generally to recognize that it is the first object in the document, so we headpatch `\@startsection`.
- Allow a tail command in `#6` to uppercase the title, so we retain DPC's braces.
- Allow each type of sectioning command to format its number differently, so we generalize `\@seccntformat`.
- Allow each type of sectioning command to format its argument differently, so we generalize `\@hangfrom`.
- Allow the starred form of the command to mark (the running head) and make an entry in the TOC, so we put `\@ssect` on the same footing as `\@sect`.

  Note that the tokens passed to the TOC now are *not* the optional argument of the command, but the required. This means that the user can no longer use the former to put variant content in to the TOC as the Manual says.

  Instead, the optional argument is used to put an alternative title into the running headers, a better choice.

\@startsection Patch a head hook into the basic sectioning command. Treat \@sect and \@ssect on an equal footing: now their pattern parts are identical.

```
\def\@startsection#1#2#3#4#5#6{%
 \@startsection@hook
 \if@noskipsec \leavevmode \fi
 \par
 \@tempskipa #4\relax
 \@afterindenttrue
 \ifdim \@tempskipa <\z@
  \@tempskipa -\@tempskipa \@afterindentfalse
 \fi
 \if@nobreak
  \everypar{}%
 \else
  \addpenalty\@secpenalty\addvspace\@tempskipa
 \fi
 \@ifstar
  {\@dblarg{\@ssect@ltx{#1}{#2}{#3}{#4}{#5}{#6}}}%
  {\@dblarg{\@sect@ltx {#1}{#2}{#3}{#4}{#5}{#6}}}%
}%
\def\@startsection@hook{}%
```

\@sect When defining \@svsec, do not expand \@seccntformat. Put brace characters back where they were before David Carlisle got at them (i.e., as if \@hangfrom had two arguments). Protect the mark mechanism from an undefined meaning. Pass #8 to the TOC instead of #7. Remove \relax from the replacement part of \@svsec.

The procedure \@hangfrom and \@runin@to can be used to process the argument of the head. The head can define, e.g., \@hangfrom@section, to do its own processing.

In using \H@refstepcounter in place of \refstepcounter we rely on either loading before any package that patches the latter, or the convention that the former is the original LaTeX procedure.

```
\class@info
  {Repairing broken LateX \string\@sect}%
\def\@sect@ltx#1#2#3#4#5#6[#7]#8{%
  \@ifnum{#2>\c@secnumdepth}{%
    \def\H@svsec{\phantomsection}%
    \let\@svsec\@empty
  }{%
    \H@refstepcounter{#1}%
    \def\H@svsec{%
     \phantomsection
    }%
    \protected@edef\@svsec{{#1}}%
    \@ifundefined{@#1cntformat}{%
     \prepdef\@svsec\@seccntformat
    }{%
```

```
    \expandafter\prepdef
    \expandafter\@svsec
                \csname @#1cntformat\endcsname
  }%
 }%
 \@tempskipa #5\relax
 \@ifdim{\@tempskipa>\z@}{%
  \begingroup
   \interlinepenalty \@M
   #6{%
    \@ifundefined{@hangfrom@#1}{\@hang@from}{\csname @hangfrom@#1\endcsname}%
    {\hskip#3\relax\H@svsec}{\@svsec}{#8}%
   }%
   \@@par
  \endgroup
  \@ifundefined{#1mark}{\@gobble}{\csname #1mark\endcsname}{#7}%
  \addcontentsline{toc}{#1}{%
   \@ifnum{#2>\c@secnumdepth}{%
    \protect\numberline{}%
   }{%
    \protect\numberline{\csname the#1\endcsname}%
   }%
   #8}%
 }{%
  \def\@svsechd{%
   #6{%
    \@ifundefined{@runin@to@#1}{\@runin@to}{\csname @runin@to@#1\endcsname}%
    {\hskip#3\relax\H@svsec}{\@svsec}{#8}%
   }%
   \@ifundefined{#1mark}{\@gobble}{\csname #1mark\endcsname}{#7}%
   \addcontentsline{toc}{#1}{%
    \@ifnum{#2>\c@secnumdepth}{%
     \protect\numberline{}%
    }{%
     \protect\numberline{\csname the#1\endcsname}%
    }%
    #8}%
  }%
 }%
 \@xsect{#5}%
}%
\def\@hang@from#1#2#3{\@hangfrom{#1#2}#3}%
\def\@runin@to #1#2#3{#1#2#3}%
```

`\@ssect` Put brace characters back where they were before David Carlisle got at them (as if `\@hangfrom` has two arguments). Possibly set a mark. Make a TOC entry.

Note that, for compatability with the `hyperref` package, we need to provide the interface required by that package (actually required by `pdfmark.def` and `nameref.sty`), namely the definition of `\@currentlabelname` (but now removed), the insertion of the procedure `\Sectionformat` (but why is this needed?), and the

call to \phantomsection (which must precede the call to \addcontentsline). We also have to sidestep the patch to \@ssect in that same file, therefore we use a different control sequence name in the call from \@startsection.

```
\def\@ssect@ltx#1#2#3#4#5#6[#7]#8{%
% \def\@currentlabelname{#8}%
  \def\H@svsec{\phantomsection}%
  \@tempskipa #5\relax
  \@ifdim{\@tempskipa>\z@}{%
    \begingroup
      \interlinepenalty \@M
      #6{%
       \@ifundefined{@hangfroms@#1}{\@hang@froms}{\csname @hangfroms@#1\endcsname}%
%      {\hskip#3\relax\H@svsec}{\Sectionformat{#8}{#1}}%
       {\hskip#3\relax\H@svsec}{#8}%
      }%
      \@@par
    \endgroup
    \@ifundefined{#1smark}{\@gobble}{\csname #1smark\endcsname}{#7}%
    \addcontentsline{toc}{#1}{\protect\numberline{}#8}%
  }{%
    \def\@svsechd{%
      #6{%
       \@ifundefined{@runin@tos@#1}{\@runin@tos}{\csname @runin@tos@#1\endcsname}%
%      {\hskip#3\relax\H@svsec}{\Sectionformat{#8}{#1}}%
       {\hskip#3\relax\H@svsec}{#8}%
      }%
      \@ifundefined{#1smark}{\@gobble}{\csname #1smark\endcsname}{#7}%
      \addcontentsline{toc}{#1}{\protect\numberline{}#8}%
    }%
  }%
  \@xsect{#5}%
}%
\def\@hang@froms#1#2{#1#2}%
\def\@runin@tos #1#2{#1#2}%
```

\init@documenthook

Document classes that incorporate this package will be hyperref-savvy. (To accomplish this, we ensure that \hyperanchor and \hyper@last are both defined.) Being hyperref-savvy levels some requirements on us, but the benefits are many.

One is that the TOC will not get amnesia and require a full set of three typesetting runs before its formatting is stable. Instead, only two runs are required: the first updates the auxiliary file, the second the TOC. However, the formatting of the document does not change.

Another aspect of being hyperref-savvy is that the syntax of commands in the ltxutil.dtx.aux file will now change if hyperref is turned on or off.

Note that \hyper@anchorstart and \hyper@anchorend constitute the programming interface for a hypertext anchor (the target of a hypertext link); \hyper@linkstart and \hyper@linkend are the interface for a hypertext link.

```
\appdef\init@documenthook{%
```

```
 \providecommand\phantomsection{}%
%\@ifx{\Sectionformat\@undefined}{\let\Sectionformat\@firstoftwo}{}%
 \providecommand\hyper@anchor[1]{}%
 \providecommand\hyper@last{}%
 \providecommand\Hy@raisedlink[1]{#1}%
 \providecommand\hyper@anchorstart[1]{}%
 \providecommand\hyper@anchorend{}%
 \providecommand\hyper@linkstart[2]{}%
 \providecommand\hyper@linkend{}%
}%
\let\H@refstepcounter\refstepcounter
```

`\sec@upcase` Upper case for sections (optional upper case items). These are created so that some headings can be toggled between mixed case and upper case readily. Headings that might be changed can be wrapped in the style file in `\sec@upcase{`⟨*text*⟩`}` constructs; the expansion of `\sec@upcase` is controlled here. It is `\relax` by default (mixed case heads), and can easily be changed to `\uppercase` if desired. If mixed-case headings are wanted by the editor, authors *must* supply mixed case text, although this is what authors should be doing anyway. (Mixed can be converted to upper, but the reverse transformation cannot be automated.)

The following setting gives the LaTeX default.

```
\def\sec@upcase#1{\relax{#1}}%
```

6.13 Patch the `tabular` and `array` Environments

`\endtabular` `\endarray` We headpatch the begin processing and tailpatch the end processing of the `tabular` and `array` environments. A document class can define these hooks as needed.

We proceed with care to make further patches to support tabulars that break over pages. Our patches will not necessarily be effective for other packages that replace the LaTeX `array` and `tabular` environments. I know of none that do so.

```
\appdef\class@documenthook{%
 \@ifpackageloaded{array}{\switch@array}{\switch@tabular}%
 \prepdef\endtabular{\endtabular@hook}%
 \@provide\endtabular@hook{}%
 \prepdef\endarray{\endarray@hook}%
 \@provide\endarray@hook{}%
 \providecommand\array@hook{}%
```

Install, effectively, a head patch to `\tabular`. In order to avoid interference from, e.g., the `array` package, we must perform this patch only *after* packages load.

```
 \prepdef\@tabular{\tabular@hook}%
 \@provide\tabular@hook{}%
}%
```

`\switch@tabular` `\switch@array` The two procedures `\switch@tabular` and `\switch@array` apply needed patches to the various tabular procedures, the former applying to the LaTeX kernel, the

latter to the required `array` package (and to the number of other required packages that load it).

```
\def\switch@tabular{%
 \let\@array@sw\@array@sw@array
 \@ifx{\@array\@array@LaTeX}{%
  \@ifx{\multicolumn\multicolumn@LaTeX}{%
   \@ifx{\@tabular\@tabular@LaTeX}{%
    \@ifx{\@tabarray\@tabarray@LaTeX}{%
     \@ifx{\array\array@LaTeX}{%
      \@ifx{\endarray\endarray@LaTeX}{%
       \@ifx{\endtabular\endtabular@LaTeX}{%
        \@ifx{\@mkpream\@mkpream@LaTeX}{%
         \@ifx{\@addamp\@addamp@LaTeX}{%
          \@ifx{\@arrayacol\@arrayacol@LaTeX}{%
           \@ifx{\@tabacol\@tabacol@LaTeX}{%
            \@ifx{\@arrayclassz\@arrayclassz@LaTeX}{%
             \@ifx{\@tabclassiv\@tabclassiv@LaTeX}{%
              \@ifx{\@arrayclassiv\@arrayclassiv@LaTeX}{%
               \@ifx{\@tabclassz\@tabclassz@LaTeX}{%
                \@ifx{\@classv\@classv@LaTeX}{%
                 \@ifx{\hline\hline@LaTeX}{%
                  \@ifx{\@tabularcr\@tabularcr@LaTeX}{%
                   \@ifx{\@xtabularcr\@xtabularcr@LaTeX}{%
                    \@ifx{\@xargarraycr\@xargarraycr@LaTeX}{%
                     \@ifx{\@yargarraycr\@yargarraycr@LaTeX}{%
                      \true@sw
                     }{%
                      \false@sw
                     }%
                    }{%
                     \false@sw
                    }%
                   }{%
                    \false@sw
                   }%
                  }{%
                   \false@sw
                  }%
                 }{%
                  \false@sw
                 }%
                }{%
                 \false@sw
                }%
               }{%
                \false@sw
               }%
              }{%
               \false@sw
              }%
             }{%
              \false@sw
             }%
            }{%
             \false@sw
```

```
              }%
             }{%
              \false@sw
             }%
            }{%
             \false@sw
            }%
           }{%
            \false@sw
           }%
          }{%
           \false@sw
          }%
         }{%
          \false@sw
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       \false@sw
      }%
     }{%
      \false@sw
     }%
    }{%
     \false@sw
    }%
   }{%
    \false@sw
   }%
  }{%
   \false@sw
  }%
}{%
 \false@sw
}%
{%
 \class@info{Patching LaTeX tabular.}%
}{%
 \class@info{Unrecognized LaTeX tabular. Please update this document class! (Proceeding with f
}%
\let\@array\@array@ltx
\let\multicolumn\multicolumn@ltx
\let\@tabular\@tabular@ltx
\let\@tabarray\@tabarray@ltx
\let\array\array@ltx
```

```
 \let\endarray\endarray@ltx
 \let\endtabular\endtabular@ltx
 \let\@mkpream\@mkpream@ltx
 \let\@addamp\@addamp@ltx
 \let\@arrayacol\@arrayacol@ltx
 \let\@tabacol\@tabacol@ltx
 \let\@arrayclassz\@arrayclassz@ltx
 \let\@tabclassiv\@tabclassiv@ltx
 \let\@arrayclassiv\@arrayclassiv@ltx
 \let\@tabclassz\@tabclassz@ltx
 \let\@classv\@classv@ltx
 \let\hline\hline@ltx
 \let\@tabularcr\@tabularcr@ltx
 \let\@xtabularcr\@xtabularcr@ltx
 \let\@xargarraycr\@xargarraycr@ltx
 \let\@yargarraycr\@yargarraycr@ltx
}%
\def\switch@array{%
 \let\@array@sw\@array@sw@LaTeX
 \@ifx{\@array\@array@array}{%
  \@ifx{\@tabular\@tabular@array}{%
   \@ifx{\@tabarray\@tabarray@array}{%
    \@ifx{\array\array@array}{%
     \@ifx{\endarray\endarray@array}{%
      \@ifx{\endtabular\endtabular@array}{%
       \@ifx{\@mkpream\@mkpream@array}{%
        \@ifx{\@classx\@classx@array}{%
         \@ifx{\insert@column\insert@column@array}{%
          \@ifx{\@arraycr\@arraycr@array}{%
           \@ifx{\@xarraycr\@xarraycr@array}{%
            \@ifx{\@xargarraycr\@xargarraycr@array}{%
             \@ifx{\@yargarraycr\@yargarraycr@array}{%
              \true@sw
         }{%
          \false@sw
         }%
         }{%
          \false@sw
         }%
         }{%
          \false@sw
         }%
         }{%
          \false@sw
         }%
         }{%
          \false@sw
         }%
        }{%
         \false@sw
```

```
        }%
       }{%
        \false@sw
       }%
      }{%
       \false@sw
      }%
     }{%
      \false@sw
     }%
    }{%
     \false@sw
    }%
   }{%
    \false@sw
   }%
  }{%
   \false@sw
  }%
 }{%
  \false@sw
 }{%
  \class@info{Patching array package.}%
 }{%
  \class@info{Unrecognized array package. Please update this document class! (Proceeding with f
 }%
 \let\@array    \@array@array@new
 \let\@@array   \@array % Cosi fan tutti
 \let\@tabular  \@tabular@array@new
 \let\@tabarray \@tabarray@array@new
 \let\array     \array@array@new
 \let\endarray  \endarray@array@new
 \let\endtabular\endtabular@array@new
 \let\@mkpream  \@mkpream@array@new
 \let\@classx   \@classx@array@new
 \let\@arrayacol\@arrayacol@ltx
 \let\@tabacol  \@tabacol@ltx
 \let\insert@column\insert@column@array@new
 \expandafter\let\csname endtabular*\endcsname\endtabular % Cosi fan tutti
 \let\@arraycr  \@arraycr@new
 \let\@xarraycr \@xarraycr@new
 \let\@xargarraycr\@xargarraycr@new
 \let\@yargarraycr\@yargarraycr@new
}%
```

`\@array@sw` The Boolean `\@array@sw` must be different depending on whether the `array` package is loaded.

```
\def\@array@sw@LaTeX{\@ifx{\\\@tabularcr}}%
\def\@array@sw@array{\@ifx{\d@llarbegin\begingroup}}%
```

`\@tabular` We provide the old versions of `\@tabular` along with the respective new versions. The change here is to avoid committing to LR mode. That will be done later (as late as possible, naturally).

```
\def\@tabular@LaTeX{%
 \leavevmode
 \hbox\bgroup$%
  \let\@acol\@tabacol
  \let\@classz\@tabclassz
  \let\@classiv\@tabclassiv
  \let\\\@tabularcr
  \@tabarray
}%
\def\@tabular@ltx{%
  \let\@acoll\@tabacoll
  \let\@acolr\@tabacolr
  \let\@acol\@tabacol
  \let\@classz\@tabclassz
  \let\@classiv\@tabclassiv
  \let\\\@tabularcr
  \@tabarray
}%
\def\@tabular@array{%
 \leavevmode
 \hbox\bgroup$%
  \col@sep\tabcolsep
  \let\d@llarbegin\begingroup
  \let\d@llarend\endgroup
  \@tabarray
}%
\def\@tabular@array@new{%
  \let\@acoll\@tabacoll
  \let\@acolr\@tabacolr
  \let\@acol\@tabacol
  \let\col@sep\@undefined
  \let\d@llarbegin\begingroup
  \let\d@llarend\endgroup
  \@tabarray
}%
```

`\@tabarray` Here we provide old and new versions of the `\@tabarray` procedure. The change here is to parametrize the default vertical alignment, which is 'c' in standard LaTeX. Under some circumstances, we want to change this to, say, 'v'.

FIXME: must decouple `array` and `tabular`.

```
\def\@tabarray@LaTeX{%
 \m@th\@ifnextchar[\@array{\@array[c]}%
}%
\def\@tabarray@ltx{%
 \m@th\@ifnextchar[\@array{\expandafter\@array\expandafter[\array@default]}%
}%
```

```
\def\@tabarray@array{%
 \@ifnextchar[{\@@array}{\@@array[c]}%
}%
\def\@tabarray@array@new{%
 \@ifnextchar[{\@@array}{\expandafter\@@array\expandafter[\array@default]}%
}%
```

\@tabularcr
\@tbpen
\@tabularcr
\@xtabularcr
\@xargarraycr
\@yargarraycr
\@arraycr
\@xarraycr

We provide for the \\ command within tabular to provide control over page breaking, just the same as that of eqnarray.

The count register \intertabularlinepenalty is similar to \interdisplaylinepenalty: it is the penalty associated with each row of a tabular. When it is set to \@M, the tabular will cleave together.

The count register \@tbpen is similar to \@eqpen: it memorizes the penalty to use after the current tabular row. If the \\ command is in its star form, then \@eqpen is set to \@M.

We append code to \samepage so that a tabular within its scope will cleave together.

We keep the standard definition of \@tabularcr in \@tabularcr@LaTeX for reference, and provide a new definition that works like \@eqncr: it sets \@tbpen to \@M if the star was given.

We also provide new versions of \@xtabularcr, \@xargarraycr, and \@yargarraycr, all of which invoke \@tbpen.

The \switch@tabular procedure switches in the new definitions.

```
\newcount\intertabularlinepenalty
\intertabularlinepenalty=100
\newcount\@tbpen
\appdef\samepage{\intertabularlinepenalty\@M}%
\def\@tabularcr@LaTeX{{\ifnum 0=`}\fi \@ifstar \@xtabularcr \@xtabularcr}%
\def\@tabularcr@ltx{{\ifnum 0=`}\fi \@ifstar {\global \@tbpen \@M \@xtabularcr }{\global \@tbpe
\def\@xtabularcr@LaTeX{\@ifnextchar [\@argtabularcr {\ifnum 0=`{\fi }\cr }}%
\def\@xtabularcr@ltx{\@ifnextchar [\@argtabularcr {\ifnum 0=`{\fi }\cr \noalign {\penalty \@tbp
\def\@xargarraycr@LaTeX#1{\@tempdima #1\advance \@tempdima \dp \@arstrutbox \vrule \@height \z@
\def\@xargarraycr@ltx#1{\@tempdima #1\advance \@tempdima \dp \@arstrutbox \vrule \@height \z@ \
\def\@yargarraycr@LaTeX#1{\cr \noalign {\vskip #1}}%
\def\@yargarraycr@ltx#1{\cr \noalign {\penalty \@tbpen \vskip #1}}%
```

If the array package has been loaded, we must alter the meanings of \@arraycr, \@xarraycr, \@xargarraycr, and \@yargarraycr. In this case, it is \switch@array that switches in the new definitions.

```
\def\@arraycr@array{%
 \relax
 \iffalse{\fi\ifnum 0=`}\fi
 \@ifstar \@xarraycr \@xarraycr
}%
\def\@arraycr@new{%
 \relax
 \iffalse{\fi\ifnum 0=`}\fi
 \@ifstar {\global \@tbpen \@M \@xarraycr }{\global \@tbpen \intertabularlinepenalty \@xarraycr
```

```
}%
\def\@xarraycr@array{%
 \@ifnextchar [%]
 \@argarraycr {\ifnum 0=‘{}\fi\cr}%
}%
\def\@xarraycr@new{%
 \@ifnextchar [%]
 \@argarraycr {\ifnum 0=‘{}\fi\cr \noalign {\penalty \@tbpen }}%
}%
\def\@xargarraycr@array#1{%
 \unskip
 \@tempdima #1\advance\@tempdima \dp\@arstrutbox
 \vrule \@depth\@tempdima \@width\z@
 \cr
}%
\def\@xargarraycr@new#1{%
 \unskip
 \@tempdima #1\advance\@tempdima \dp\@arstrutbox
 \vrule \@depth\@tempdima \@width\z@
 \cr
 \noalign {\penalty \@tbpen }%
}%
\def\@yargarraycr@array#1{%
 \cr
 \noalign{\vskip #1}%
}%
\def\@yargarraycr@new#1{%
 \cr
 \noalign{\penalty \@tbpen \vskip #1}%
}%
```

`\array` We provide old and new versions of the `\array` procedure for both LaTeX and the `array` package. The change here is to accomodate the new procedures that will be called for the array boundaries, even though at present they are not special. A thought: here is where matrices can be readily accomodated.

```
\def\array@LaTeX{%
 \let\@acol\@arrayacol
 \let\@classz\@arrayclassz
 \let\@classiv\@arrayclassiv
 \let\\\@arraycr
 \let\@halignto\@empty
 \@tabarray
}%
\def\array@ltx{%
 \@ifmmode{}{\@badmath$}%
 \let\@acoll\@arrayacol
 \let\@acolr\@arrayacol
 \let\@acol\@arrayacol
 \let\@classz\@arrayclassz
 \let\@classiv\@arrayclassiv
```

```
 \let\\\@arraycr
 \let\@halignto\@empty
 \@tabarray
}%
\def\array@array{%
 \col@sep\arraycolsep
 \def\d@llarbegin{$}\let\d@llarend\d@llarbegin\gdef\@halignto{}%
 \@tabarray
}
\def\array@array@new{%
 \@ifmmode{}{\@badmath$}%
 \let\@acoll\@arrayacol
 \let\@acolr\@arrayacol
 \let\@acol\@arrayacol
\let\col@sep\@undefined
 \def\d@llarbegin{$}%
 \let\d@llarend\d@llarbegin
 \gdef\@halignto{}%
 \@tabarray
}%
```

`\@array` Here we provide old and new versions of `\@array`. The change here is to provide a convenient, flexible, and extensible mechanism for new vertical alignment options.

Instead of testing the optional argument with `\if`, we use a dispatcher based on `\csname`.

We also refrain from using `\ialign`, which would set the `\tabskip` to the wrong value.

Finally, the procedure to set the `\@arstrutbox` is broken out so that it can be patched.

```
\def\@array@LaTeX[#1]#2{%
  \if #1t\vtop \else \if#1b\vbox \else \vcenter \fi\fi
  \bgroup
  \setbox\@arstrutbox\hbox{%
    \vrule \@height\arraystretch\ht\strutbox
           \@depth\arraystretch \dp\strutbox
           \@width\z@}%
  \@mkpream{#2}%
  \edef\@preamble{%
    \ialign \noexpand\@halignto
      \bgroup \@arstrut \@preamble \tabskip\z@skip \cr}%
  \let\@startpbox\@@startpbox \let\@endpbox\@@endpbox
  \let\tabularnewline\\%
    \let\par\@empty
    \let\@sharp##%
    \set@typeset@protect
    \lineskip\z@skip\baselineskip\z@skip
    \ifhmode \@preamerr\z@ \@@par\fi
    \@preamble
}%
```

```
\def\@array@ltx[#1]#2{%
 \@nameuse{@array@align@#1}%
  \set@arstrutbox
  \@mkpream{#2}%
  \prepdef\@preamble{%
    \tabskip\tabmid@skip
    \@arstrut
  }%
  \appdef\@preamble{%
    \tabskip\tabright@skip
    \cr
    \array@row@pre
  }%
% \let\@startpbox\@@startpbox
% \let\@endpbox\@@endpbox
  \let\tabularnewline\\%
  \let\par\@empty
  \let\@sharp##%
  \set@typeset@protect
  \lineskip\z@skip\baselineskip\z@skip
  \tabskip\tableft@skip\relax
  \ifhmode \@preamerr\z@ \@@par\fi
  \everycr{}%
  \expandafter\halign\expandafter\@halignto\expandafter\bgroup\@preamble
}%
%
\def\set@arstrutbox{%
  \setbox\@arstrutbox\hbox{%
    \vrule \@height\arraystretch\ht\strutbox
           \@depth\arraystretch \dp\strutbox
           \@width\z@
  }%
}%
```

\@array@array

```
\def\@array@array[#1]#2{%
  \@tempdima \ht \strutbox
  \advance \@tempdima by\extrarowheight
  \setbox \@arstrutbox \hbox{\vrule
             \@height \arraystretch \@tempdima
             \@depth \arraystretch \dp \strutbox
             \@width \z@}%
  \begingroup
  \@mkpream{#2}%
  \xdef\@preamble{\noexpand \ialign \@halignto
                  \bgroup \@arstrut \@preamble
                          \tabskip \z@ \cr}%
  \endgroup
  \@arrayleft
  \if #1t\vtop \else \if#1b\vbox \else \vcenter \fi \fi
```

```
  \bgroup
  \let \@sharp ##\let \protect \relax
  \lineskip \z@
  \baselineskip \z@
  \m@th
  \let\\\@arraycr \let\tabularnewline\\\let\par\@empty \@preamble
}%
\def\@array@array@new[#1]#2{%
  \@tempdima\ht\strutbox
  \advance\@tempdima by\extrarowheight
  \setbox\@arstrutbox\hbox{%
   \vrule \@height\arraystretch\@tempdima
          \@depth \arraystretch\dp\strutbox
          \@width \z@
  }%
  \begingroup
   \@mkpream{#2}%
   \xdef\@preamble{\@preamble}%
  \endgroup
  \prepdef\@preamble{%
   \tabskip\tabmid@skip
    \@arstrut
  }%
  \appdef\@preamble{%
   \tabskip\tabright@skip
   \cr
   \array@row@pre
  }%
  \@arrayleft
  \@nameuse{@array@align@#1}%
  \m@th
  \let\\\@arraycr
  \let\tabularnewline\\%
  \let\par\@empty
  \let\@sharp##%
  \set@typeset@protect
  \lineskip\z@\baselineskip\z@
  \tabskip\tableft@skip
  \everycr{}%
  \expandafter\halign\expandafter\@halignto\expandafter\bgroup\@preamble
}%
```

`\endarray` Here we provide old and new versions of `\endarray`. The change here is to use a single procedure to close out any array-like structure, namely `\endarray@ltx`. It merely closes out the `\halign`.

```
\def\endarray@LaTeX{%
 \crcr\egroup\egroup
}%
\def\endarray@ltx{%
 \crcr\array@row@pst\egroup\egroup
```

```
}%
\def\endarray@array{%
 \crcr \egroup \egroup \@arrayright \gdef\@preamble{}%
}%
\def\endarray@array@new{%
 \crcr\array@row@pst\egroup\egroup % Same as \endarray@ltx
 \@arrayright
 \global\let\@preamble\@empty
}%
```

`\endtabular`

```
\def\endtabular@LaTeX{%
 \crcr\egroup\egroup $\egroup
}%
\def\endtabular@ltx{%
 \endarray
}%
\def\endtabular@array{%
 \endarray $\egroup
}%
\def\endtabular@array@new{%
 \endarray
}%
```

`endtabular*` Here we provide a proper definition for the star-form of `\end{endtabular}`. It is one of the enduring curiosities that the LaTeX kernel continues to use dangerously and inappropriately "optimized" definitions for such commands.

```
\@namedef{endtabular*}{\endtabular}%
```

`\multicolumn`

```
\long\def\multicolumn@LaTeX#1#2#3{%
 \multispan{#1}\begingroup
  \@mkpream{#2}%
  \def\@sharp{#3}\set@typeset@protect
  \let\@startpbox\@@startpbox\let\@endpbox\@@endpbox
  \@arstrut \@preamble\hbox{}\endgroup\ignorespaces
}%
\long\def\multicolumn@ltx#1#2#3{%
 \multispan{#1}%
 \begingroup
  \@mkpream{#2}%
  \def\@sharp{#3}%
  \set@typeset@protect
 %\let\@startpbox\@@startpbox\let\@endpbox\@@endpbox
  \@arstrut
  \@preamble
  \hbox{}%
 \endgroup
 \ignorespaces
}%
```

\@array@align@
\array@default

Here are the various procedures for the vertical alignment options. The change from standard LaTeX is that we do not go into math mode in every case: only when required by \vcenter. Also, we use \aftergroup to close out the boxes and modes we have started. It requires only that each procedure issue exactly one unmatched \bgroup.

We establish here the default vertical alignment.

```
\def\@array@align@t{\leavevmode\vtop\bgroup}%
\def\@array@align@b{\leavevmode\vbox\bgroup}%
\def\@array@align@c{\leavevmode\@ifmmode{\vcenter\bgroup}{$\vcenter\bgroup\aftergroup$\aftergro
\def\@array@align@v{%
 \@ifmmode{%
  \@badmath
  \vcenter\bgroup
 }{%
  \@ifinner{%
   $\vcenter\bgroup\aftergroup$
  }{%
   \@@par\bgroup
  }%
 }%
}%
\def\array@default{c}%
```

\array@row@pre
\array@row@pst
\array@row@rst

The procedure \array@row@rst reestablishes a default context for an alignment, so that they can be nested. Any environment or procedure that alters the way alignments are formatted must patch this procedure to restore from that alteration. To start things off, we equate \@array@align@v to \@array@align@c, because it does not make sense to do the former in any context other than the MVL or in a list that will be unboxed onto the MVL.

```
\def\array@row@rst{%
 \let\@array@align@v\@array@align@c
}%
\def\array@row@pre{}%
\def\array@row@pst{}%
```

\toprule
\colrule
\botrule

Default definitions for \toprule, \colrule, \botrule

```
\newcommand\toprule{\tab@rule{\column@font}{\column@fil}{\frstrut}}%
\newcommand\colrule{\unskip\lrstrut\\\tab@rule{\body@font}{}{\frstrut}}%
\newcommand\botrule{\unskip\lrstrut\\\noalign{\hline@rule}{}}%
```

\hline

```
\def\hline@LaTeX{%
 \noalign{\ifnum0=`}\fi\hrule \@height \arrayrulewidth \futurelet
   \reserved@a\@xhline
}%
\def\hline@ltx{%
 \noalign{%
  \ifnum0=`}\fi
```

```
  \hline@rule
  \futurelet\reserved@a\@xhline
 % \noalign ended in \@xhline
}%
\def\@xhline@unneeded{%
 \say\reserved@a
 \ifx\reserved@a\hline
  \vskip\doublerulesep
  \vskip-\arrayrulewidth
 \fi
 \ifnum0=`{\fi}%
}%
\def\tab@rule#1#2#3{%
 \crcr
 \noalign{%
  \hline@rule
  \gdef\@arstrut@hook{%
   \global\let\@arstrut@hook\@empty
   #3%
  }%
  \gdef\cell@font{#1}%
  \gdef\cell@fil{#2}%
 }%
}%
\def\column@font{}%
\def\column@fil{}%
\def\body@font{}%
\def\cell@font{}%
\def\frstrut{}%
\def\lrstrut{}%
```

`\@arstrut@hline` `\@arstrut@org` `\@arstrut@hook` `\@arstrutbox@hline` `\set@arstrutbox` `\hline@rule`

The procedure `\@arstrut@hline` is substantially the same as `\@arstrut`, except the strut copied in is `\@arstrutbox@hline`instead of `\@arstrutbox`.

The procedure `\@arstrut@hook` is redefined in `\tab@rule`!

The register `\@arstrutbox@hline`.

We append to `\set@arstrutbox` the code necessary to set a strut following an `\hline`.

The procedure `\hline@rule` lays down a rule, and changes the meaning of `\@arstrut` so that the next line will be correctly strutted.

The `\@arstrut@hline@clnc` is a klootch, a magic number.

```
\def\@arstrut@hline{%
 \relax
 \@ifmmode{\copy}{\unhcopy}\@arstrutbox@hline
 \@arstrut@hook
}%
%
\let\@arstrut@org\@arstrut
\def\@arstrut@hook{%
 \global\let\@arstrut\@arstrut@org
```

```
}%
%
\newbox\@arstrutbox@hline
\appdef\set@arstrutbox{%
  \setbox\@arstrutbox@hline\hbox{%
    \setbox\z@\hbox{$0^{0}_{}$}%
    \dimen@\ht\z@\advance\dimen@\@arstrut@hline@clnc
    \@ifdim{\dimen@<\arraystretch\ht\strutbox}{\dimen@=\arraystretch\ht\strutbox}{}%
    \vrule \@height\dimen@
           \@depth\arraystretch \dp\strutbox
           \@width\z@
  }%
}%
%
\def\hline@rule{%
 \hrule \@height \arrayrulewidth
 \global\let\@arstrut\@arstrut@hline
}%
\def\@arstrut@hline@clnc{2\p@}% % Klootch: magic number
```

`\tableft@skip`

```
\def\tableft@skip{\z@skip}%
\def\tabmid@skip{\z@skip}%\@flushglue
\def\tabright@skip{\z@skip}%
\def\tableftsep{\tabcolsep}%
\def\tabmidsep{\tabcolsep}%
\def\tabrightsep{\tabcolsep}%
\def\cell@fil{}%
\def\pbox@hook{}%
```

`\@arstrut`

```
\appdef\@arstrut{\@arstrut@hook}%
\let\@arstrut@hook\@empty
\def\@addtopreamble{\appdef\@preamble}%
```

`\@mkpream`

```
\def\@mkpream@LaTeX#1{%
  \@firstamptrue\@lastchclass6
  \let\@preamble\@empty
  \let\protect\@unexpandable@protect
  \let\@sharp\relax
  \let\@startpbox\relax\let\@endpbox\relax
  \@expast{#1}%
  \expandafter\@tfor \expandafter
    \@nextchar \expandafter:\expandafter=\reserved@a\do
       {\@testpach\@nextchar
    \ifcase \@chclass \@classz \or \@classi \or \@classii \or \@classiii
      \or \@classiv \or\@classv \fi\@lastchclass\@chclass}%
  \ifcase \@lastchclass \@acol
```

```
      \or \or \@preamerr \@ne\or \@preamerr \tw@\or \or \@acol \fi
}%
\def\@mkpream@ltx#1{%
 \@firstamptrue
 \@lastchclass6
 \let\@preamble\@empty
 \let\protect\@unexpandable@protect
 \let\@sharp\relax
%\let\@startpbox\relax\let\@endpbox\relax
 \@expast{#1}%
 \expandafter\@tfor\expandafter\@nextchar\expandafter:\expandafter=\reserved@a
 \do{%
  \expandafter\@testpach\expandafter{\@nextchar}%
  \ifcase\@chclass
   \@classz
  \or
   \@classi
  \or
   \@classii
  \or
   \@classiii
  \or
   \@classiv
  \or
   \@classv
  \fi
  \@lastchclass\@chclass
 }%
 \ifcase\@lastchclass
  \@acolr % right-hand column
 \or
 \or
  \@preamerr\@ne
 \or
  \@preamerr\tw@
 \or
 \or
  \@acolr % right-hand column
 \fi
}%
```

\insert@column

```
\def\insert@column@array{%
   \the@toks \the \@tempcnta
   \ignorespaces \@sharp \unskip
   \the@toks \the \count@ \relax
}%
\def\insert@column@array@new{%
 \the@toks\the\@tempcnta
 \array@row@rst\cell@font
```

```
\ignorespaces\@sharp\unskip
\the@toks\the\count@
\relax
}%
```

`\@mkpream@relax` The procedure `\@mkpream@relax` participates in a strange and wonderful method of binding the alignment procedure—but only certain parts thereof.

Here is how it works: in LaTeX, the `array` package, and in the `longtable` package alike, there is a need to create an alignment preamble (using `\@mkpream`) for use by the upcoming `\halign`. Then, in both `array` and `longtable`, TeX's `\edef` is used to 'compile in place' that alignment preamble.

In the case of `array`, the operation is done in order to pre-expand the use of `*`, in `longtable`, it is to set the widths of the columns.

Now, during this `\edef`, certain control sequence names must *not* be expanded, and those are robustified by `\@mkpream@relax`.

```
\def\@mkpream@relax{%
 \let\tableftsep\relax
 \let\tabmidsep\relax
 \let\tabrightsep\relax
 \let\array@row@rst\relax
 \let\cell@font\relax
 \let\@startpbox\relax
}%
```

`\@mkpream`

```
\def\@mkpream@array#1{%
   \gdef\@preamble{}\@lastchclass 4 \@firstamptrue
   \let\@sharp\relax \let\@startpbox\relax \let\@endpbox\relax
   \@temptokena{#1}\@tempswatrue
   \@whilesw\if@tempswa\fi{\@tempswafalse\the\NC@list}%
   \count@\m@ne
   \let\the@toks\relax
   \prepnext@tok
   \expandafter \@tfor \expandafter \@nextchar
    \expandafter :\expandafter =\the\@temptokena \do
   {\@testpach
   \ifcase \@chclass \@classz \or \@classi \or \@classii
     \or \save@decl \or \or \@classv \or \@classvi
     \or \@classvii \or \@classviii
     \or \@classx
     \or \@classx \fi
   \@lastchclass\@chclass}%
   \ifcase\@lastchclass
   \@acol \or
   \or
   \@acol \or
   \@preamerr \thr@@ \or
   \@preamerr \tw@ \@addtopreamble\@sharp \or
   \or
```

```
   \else  \@preamerr \@ne \fi
   \def\the@toks{\the\toks}%
}%
\def\@mkpream@array@new#1{%
 \gdef\@preamble{}%
 \@lastchclass\f@ur
 \@firstamptrue
 \let\@sharp\relax
 \@mkpream@relax
%\let\@startpbox\relax\let\@endpbox\relax
 \@temptokena{#1}\@tempswatrue
 \@whilesw\if@tempswa\fi{\@tempswafalse\the\NC@list}%
 \count@\m@ne
 \let\the@toks\relax
 \prepnext@tok
 \expandafter\@tfor\expandafter\@nextchar\expandafter:\expandafter=\the\@temptokena
 \do{%
  \@testpach
  \ifcase\@chclass
   \@classz
  \or
   \@classi
  \or
   \@classii
  \or
   \save@decl
  \or
  \or
   \@classv
  \or
   \@classvi
  \or
   \@classvii
  \or
   \@classviii
  \or
   \@classx
  \or
   \@classx
  \fi
  \@lastchclass\@chclass
 }%
 \ifcase\@lastchclass
  \@acolr % right-hand column
 \or
 \or
  \@acolr % right-hand column
 \or
  \@preamerr\thr@@
 \or
```

```
   \@preamerr\tw@\@addtopreamble\@sharp
  \or
  \or
  \else
   \@preamerr\@ne
  \fi
  \def\the@toks{\the\toks}%
}%
```

\@addamp

```
\def\@addamp@LaTeX{%
   \if@firstamp\@firstampfalse\else\edef\@preamble{\@preamble &}\fi
}%
\def\@addamp@ltx{%
  \if@firstamp\@firstampfalse\else\@addtopreamble{&}\fi
}%
```

\@arrayacol

```
\def\@arrayacol@LaTeX{%
  \edef\@preamble{\@preamble \hskip \arraycolsep}%
}%
\def\@arrayacol@ltx{%
  \@addtopreamble{\hskip\arraycolsep}%
}%
```

\@tabacol

```
\def\@tabacoll{%
  \@addtopreamble{\hskip\tableftsep\relax}%
}%
\def\@tabacol@LaTeX{%
  \edef\@preamble{\@preamble \hskip \tabcolsep}%
}%
\def\@tabacol@ltx{%
  \@addtopreamble{\hskip\tabmidsep\relax}%
}%
\def\@tabacolr{%
  \@addtopreamble{\hskip\tabrightsep\relax}%
}%
```

\@arrayclassz

```
\def\@arrayclassz@LaTeX{%
  \ifcase \@lastchclass \@acolampacol \or \@ampacol \or
    \or \or \@addamp \or
    \@acolampacol \or \@firstampfalse \@acol \fi
  \edef\@preamble{\@preamble
   \ifcase \@chnum
      \hfil$\relax\@sharp$\hfil \or $\relax\@sharp$\hfil
     \or \hfil$\relax\@sharp$\fi}%
}%
```

```
\def\@arrayclassz@ltx{%
 \ifcase\@lastchclass
  \@acolampacol
 \or
  \@ampacol
 \or
 \or
 \or
  \@addamp
 \or
  \@acolampacol
 \or
  \@firstampfalse\@acoll
 \fi
 \ifcase\@chnum
  \@addtopreamble{%
   \hfil\array@row@rst$\relax\@sharp$\hfil
  }%
 \or
  \@addtopreamble{%
   \array@row@rst$\relax\@sharp$\hfil
  }%
 \or
  \@addtopreamble{%
   \hfil\array@row@rst$\relax\@sharp$%
  }%
 \fi
}%
```

\@tabclassz

```
\def\@tabclassz@LaTeX{%
  \ifcase\@lastchclass
    \@acolampacol
  \or
    \@ampacol
  \or
  \or
  \or
    \@addamp
  \or
    \@acolampacol
  \or
    \@firstampfalse\@acol
  \fi
  \edef\@preamble{%
    \@preamble{%
      \ifcase\@chnum
        \hfil\ignorespaces\@sharp\unskip\hfil
      \or
        \hskip1sp\ignorespaces\@sharp\unskip\hfil
```

```
      \or
        \hfil\hskip1sp\ignorespaces\@sharp\unskip
      \fi}}%
}%
\def\@tabclassz@ltx{%
 \ifcase\@lastchclass
  \@acolampacol
 \or
  \@ampacol
 \or
 \or
 \or
  \@addamp
 \or
  \@acolampacol
 \or
  \@firstampfalse\@acoll
 \fi
 \ifcase\@chnum
  \@addtopreamble{%
   {\hfil\array@row@rst\cell@font\ignorespaces\@sharp\unskip\hfil}%
  }%
 \or
  \@addtopreamble{%
   {\cell@fil\hskip1sp\array@row@rst\cell@font\ignorespaces\@sharp\unskip\hfil}%
  }%
 \or
  \@addtopreamble{%
   {\hfil\hskip1sp\array@row@rst\cell@font\ignorespaces\@sharp\unskip\cell@fil}%
  }%
 \fi
}%
```

\@tabclassiv

```
\def\@tabclassiv@LaTeX{%
 \@addtopreamble\@nextchar
}%
\def\@tabclassiv@ltx{%
 \expandafter\@addtopreamble\expandafter{\@nextchar}%
}%
```

\@arrayclassiv

```
\def\@arrayclassiv@LaTeX{%
 \@addtopreamble{$\@nextchar$}%
}%
\def\@arrayclassiv@ltx{%
 \expandafter\@addtopreamble\expandafter{\expandafter$\@nextchar$}%
}%
```

\@classv

```
\def\@classv@LaTeX{%
 \@addtopreamble{\@startpbox{\@nextchar}\ignorespaces
 \@sharp\@endpbox}%
}%
\def\@classv@ltx{%
 \expandafter\@addtopreamble
 \expandafter{%
 \expandafter \@startpbox
 \expandafter {\@nextchar}%
 \pbox@hook\array@row@rst\cell@font\ignorespaces\@sharp\@endpbox
 }%
}%
```

\@classx

```
\def\@classx@array{%
  \ifcase \@lastchclass
  \@acolampacol \or
  \@addamp \@acol \or
  \@acolampacol \or
  \or
  \@acol \@firstampfalse \or
  \@addamp
  \fi
}%
\def\@classx@array@new{%
 \ifcase \@lastchclass
  \@acolampacol
 \or
  \@addamp \@acol
 \or
  \@acolampacol
 \or
 \or
  \@firstampfalse\@acoll
 \or
  \@addamp
 \fi
}%
```

6.14 Repair other broken parts of LaTeX

\@xbitor Expansion part has extraneous space token. Removed.

```
\def\@xbitor@LaTeX #1{\@tempcntb \count#1
   \ifnum \@tempcnta =\z@
   \else
     \divide\@tempcntb\@tempcnta
     \ifodd\@tempcntb \@testtrue\fi
   \fi}%
\def\@xbitor@ltx#1{%
```

```
  \@tempcntb\count#1%
  \@ifnum{\@tempcnta=\z@}{}{%
   \divide\@tempcntb\@tempcnta
   \@ifodd\@tempcntb{\@testtrue}{}%
  }%
 }%
 \@ifx{\@xbitor\@xbitor@LaTeX}{%
   \class@info{Repairing broken LaTeX \string\@xbitor}%
 }{%
   \class@info{Unrecognized LaTeX \string\@xbitor. Please update this document class! (Proceedin
 }%
 \let\@xbitor\@xbitor@ltx
```

6.15 Syntax

`\@gobble@opt@one` The `\@gobble@opt@one` command eats up an optional argument and one required argument.

```
\newcommand*\@gobble@opt@one[2][]{}%
```

6.16 Auto-indented Contents

Facility to automatically determine the proper indentation of the TOC entries.

Note on `hyperref` compatibility: We must respect that `\contentsline`now has a 4th argument. So, instead of trying to override the meaning of `\contentsline`, we use the aux file to remember max values from one run to the next.

In this respect, this package retains compatability with `hyperref`.

`\@starttoc` Install hooks at beginning and end of the TOC processing.

```
\def\@starttoc#1{%
  \begingroup
    \toc@pre
    \makeatletter
    \@input{\jobname.#1}%
    \if@filesw
      \expandafter\newwrite\csname tf@#1\endcsname
      \immediate\openout \csname tf@#1\endcsname \jobname.#1\relax
    \fi
    \@nobreakfalse
    \toc@post
  \endgroup
}%
\def\toc@pre{}%
\def\toc@post{}%
```

`\toc@@font` Interface for setting the formatting characteristics of this part of the TOC.

Note: `\toc@@font` is the common font for all auto-sizing toc commands, although this, too, could become a dispatcher.

```
\def\toc@@font{}%{\footnotesize\rmfamily}%
\def\@dotsep{\z@}%{5.5pt}%
```

`\l@section` Interface for determining which TOC elements are automatically indented.

All of the `\l@...` commands simply go through the bottleproc `\l@@sections`. The calling convention is to pass the name of self and the name of parent. If you want to exclude any of these from the indentation scheme, simply leave the `\l@...` command undefined.

Note that the parent of "section" is nil, so we have to define a stub.

```
%\def\l@section{%
% \l@@sections{}{section}% Implicit #3#4
%}%
%\def\tocleft@{\z@}%
%\def\l@subsection{%
% \l@@sections{section}{subsection}% Implicit #3#4
%}%
%\def\l@subsubsection{%
% \l@@sections{subsection}{subsubsection}% Implicit #3#4
%}%
%\def\l@paragraph{%
% \l@@sections{subsubsection}{paragraph}% Implicit #3#4
%}%
%\def\l@subparagraph#1#2{%
% \l@@sections{paragraph}{subparagraph}% Implicit #3#4
%}%
%
```

Glom some `\dimen` registers.

```
\let\tocdim@section       \leftmargini
\let\tocdim@subsection    \leftmarginii
\let\tocdim@subsubsection \leftmarginiii
\let\tocdim@paragraph     \leftmarginiv
\let\tocdim@appendix      \leftmarginv
\let\tocdim@pagenum       \leftmarginvi
```

`\toc@pre@auto` `\toc@post@auto` We patch `\@starttoc` to: 1) before TOC processing, initialize the max registers and set the needed dimensions from the values stored in the auxiliary file, and 2) after TOC processing, store out those max register values into the auxiliary file.

Note that the font is set here: all other TOC entries must override these font settings.

To activate this override of the standard LaTeX processing, the substyle does: `\let\toc@pre\toc@pre@auto` and `\let\toc@post\toc@post@auto`.

```
\def\toc@pre@auto{%
  \toc@@font
  \@tempdima\z@
  \toc@setindent\@tempdima{section}%
  \toc@setindent\@tempdima{subsection}%
  \toc@setindent\@tempdima{subsubsection}%
  \toc@setindent\@tempdima{paragraph}%
  \toc@letdimen{appendix}%
  \toc@letdimen{pagenum}%
```

```
}%
\def\toc@post@auto{%
  \if@filesw
   \begingroup
    \toc@writedimen{section}%
    \toc@writedimen{subsection}%
    \toc@writedimen{subsubsection}%
    \toc@writedimen{paragraph}%
    \toc@writedimen{appendix}%
    \toc@writedimen{pagenum}%
   \endgroup
  \fi
}%
```

`\toc@setindent`

```
\def\toc@setindent#1#2{%
 \csname tocdim@#2\endcsname\tocdim@min\relax
 \@ifundefined{tocmax@#2}{\@namedef{tocmax@#2}{\z@}}{}%
 \advance#1\@nameuse{tocmax@#2}\relax
 \expandafter\edef\csname tocleft@#2\endcsname{\the#1}%
}%
```

`\toc@letdimen`

```
\def\toc@letdimen#1{%
 \csname tocdim@#1\endcsname\tocdim@min\relax
 \@ifundefined{tocmax@#1}{\@namedef{tocmax@#1}{\z@}}{}%
 \expandafter\let\csname tocleft@#1\expandafter\endcsname\csname tocmax@#1\endcsname
}%
```

`\toc@writedimen`

```
\def\toc@writedimen#1{%
 \immediate\write\@auxout{%
  \gdef\expandafter\string\csname tocmax@#1\endcsname{%
   \expandafter\the\csname tocdim@#1\endcsname
  }%
 }%
}%
```

`\l@@sections` The procedure for formatting the indented TOC entries. We use control sequence names such as `\tocmax@section` and `\tocleft@section`, the former being written to the auxiliary file and the latter only defined for the duration of the TOC processing.

Note that the assignment of `\box\z@` must endure over the invocation of `#3`.

```
\def\l@@sections#1#2#3#4{%
 % #1 - superior section
 % #2 - this section
 % #3 - content, including possible \numberline
 % #4 - page number
 \begingroup
```

```
  \everypar{}%
  \set@tocdim@pagenum{#4}%
  \global\@tempdima\csname tocdim@#2\endcsname
  \leftskip\csname tocleft@#2\endcsname\relax
  \dimen@\csname tocleft@#1\endcsname\relax
  \parindent-\leftskip\advance\parindent\dimen@
  \rightskip\tocleft@pagenum plus 1fil\relax
  \skip@\parfillskip\parfillskip\z@
  \let\numberline\numberline@@sections
  \@nameuse{l@f@#2}%
  \ignorespaces#3\unskip\nobreak\hskip\skip@
  \hb@xt@\rightskip{\hfil\unhbox\@tempboxa}\hskip-\rightskip\hskip\z@skip
  \par
  \expandafter\aftergroup\csname tocdim@#2\endcsname\expandafter
 \endgroup\the\@tempdima\relax
}%
\def\set@tocdim@pagenum#1{%
 \setbox\@tempboxa\hbox{\ignorespaces#1}%
 \@ifdim{\tocdim@pagenum<\wd\z@}{\global\tocdim@pagenum\wd\z@}{}%
}%
```

`\numberline@@sections` The bottleproc for all `\numberline` processing in indented TOC entries. The first argument is self.

We use `\@tempdima` to pass a value around (via global assignment) because `\numberline` executes inside a group if the `hyperref` package is loaded. Would that it were not so!

```
\def\numberline@@sections#1{%
 \leavevmode\hb@xt@-\parindent{%
  \hfil
  \@if@empty{#1}{}{%
   \setbox\z@\hbox{#1.\kern\@dotsep}%
   \@ifdim{\@tempdima<\wd\z@}{\global\@tempdima\wd\z@}{}%
   \unhbox\z@
  }%
 }%
 \ignorespaces
}%
\def\tocdim@min{\z@}%
```

6.17 Lists

`\list` Using `\parshape` to implement lists was always suspect (can you get behind `\parshape\@ne`?) and we now see that it was a mistake all along. Why? Because `\parshape`, like `\hangindent`, achieves its effect via "shifting" the `\hbox`es in a paragraph instead of using `\leftskip` and `\parindent`, which is robust during column balancing.

We introduce the alternative method with a hook into the LaTeX kernel procedure `\list`, which is the implementation of all lists.

```
\def\list#1#2{%
  \ifnum \@listdepth >5\relax
    \@toodeep
  \else
    \global\advance\@listdepth\@ne
  \fi
  \rightmargin\z@
  \listparindent\z@
  \itemindent\z@
  \csname @list\romannumeral\the\@listdepth\endcsname
  \def\@itemlabel{#1}%
  \let\makelabel\@mklab
  \@nmbrlistfalse
  #2\relax
  \@trivlist
  \parskip\parsep
  \set@listindent
  \ignorespaces
}%
\def\set@listindent@parshape{%
 \parindent\listparindent
 \advance\@totalleftmargin\leftmargin
 \advance\linewidth-\rightmargin
 \advance\linewidth-\leftmargin
 \parshape\@ne\@totalleftmargin\linewidth
}%
\def\set@listindent@{%
 \parindent\listparindent
 \advance\@totalleftmargin\leftmargin
 \advance\rightskip\rightmargin
 \advance\leftskip\@totalleftmargin
}%
\let\set@listindent\set@listindent@parshape
```

6.18 End of the `ltxutil` DOCSTRIP module

Here ends the module.

```
%</ltxutil-krn>
```

Here ends the programmer's documentation.

Index

Symbols

\% 711
\(. 527
\) 528
.dtx 1, 4–6
.ins 4
.tfm 1
.vf 1
.vpl 1
\@@array 1041, 1102, 1105
\@@end 340
\@@endpbox 1204, 1227, 1333, 1342
\@@par . 661, 783, 825, 1210, 1234, 1360
\@@startpbox . 1204, 1226, 1333, 1342
\@Esphack 699, 701
\@M 40
\@MM 500
\@acol 1063, 1072, 1089, 1159, 1170, 1186, 1461, 1462, 1540, 1542, 1631, 1677, 1744, 1747, 1755
\@acolampacol 1629, 1631, 1639, 1647, 1667, 1675, 1691, 1699, 1743, 1745, 1753, 1757
\@acoll . . 1070, 1087, 1168, 1184, 1649, 1701, 1760
\@acolr . . 1071, 1088, 1169, 1185, 1491, 1499, 1589, 1592
\@addamp 877, 967, 1604, 1630, 1645, 1673, 1697, 1744, 1748, 1755, 1762
\@addamp@LaTeX 877, 1604
\@addamp@ltx 967, 1607
\@addtopreamble 1448, 1544, 1596, 1608, 1614, 1617, 1623, 1626, 1652, 1656, 1660, 1704, 1708, 1712, 1718, 1721, 1724, 1727, 1730, 1734
\@afterindentfalse 743
\@afterindenttrue 741
\@ampacol 1629, 1641, 1669, 1693
\@argarraycr 1131, 1135
\@argswap 13
\@argswap 255
\@argswap@val 13
\@argswap@val 255
\@argtabularcr 1113, 1114
\@array 42
\@array 869, 959, 983, 1040, 1041, 1096, 1099, 1193
\@array@LaTeX 869, 1193
\@array@align@ 1349
\@array@align@b 1350
\@array@align@c 46
\@array@align@c 1351, 1366
\@array@align@t 1349
\@array@align@v 46
\@array@align@v 1352, 1366
\@array@array 983, 1246
\@array@array@new . . 1040, 1268
\@array@ltx 959, 1213
\@array@sw 38
\@array@sw 868, 982, 1058
\@array@sw@LaTeX 982, 1058
\@array@sw@array 868, 1059
\@arrayacol 878, 968, 1049, 1159, 1168–1170, 1184–1186, 1610
\@arrayacol@LaTeX . . . 878, 1610
\@arrayacol@ltx 968, 1049, 1613
\@arrayclassiv . . 882, 972, 1161, 1172, 1723
\@arrayclassiv@LaTeX 882, 1723
\@arrayclassiv@ltx . . 972, 1726
\@arrayclassz . . . 880, 970, 1160, 1171, 1628
\@arrayclassz@LaTeX . 880, 1628
\@arrayclassz@ltx . . . 970, 1637
\@arraycr 40
\@arraycr . 992, 1053, 1107, 1162, 1173, 1266, 1292
\@arraycr@array 992, 1119
\@arraycr@new 1053, 1124

\@arrayleft 1259, 1289
\@arrayright 1309, 1313
\@arstrut 47
\@arstrut 1203, 1219, 1256, 1282, 1334, 1343, 1416, 1418, 1435, 1446
\@arstrut@hline 47
\@arstrut@hline 1410
\@arstrut@hline@clnc 47
\@arstrut@hline@clnc 1425, 1437
\@arstrut@hook 47
\@arstrut@hook 1396, 1397, 1410, 1446, 1447
\@arstrut@org 1410
\@arstrutbox 42, 47
\@arstrutbox 1115, 1116, 1139, 1145, 1196, 1240, 1249, 1271
\@arstrutbox@hline 47
\@arstrutbox@hline 1410
\@auxout 525, 542, 1842
\@boole@def 13
\@boole@def 265, 277–280, 282–292
\@boolean 13
\@boolean 265
\@booleanfalse 14
\@booleanfalse 275, 607, 608, 666
\@booleantrue 14
\@booleantrue 275, 590, 638
\@bsphack 541, 704
\@caption 25
\@captype 610
\@centering . 413, 416, 419, 438, 444, 457
\@chclass 1459, 1460, 1475, 1488, 1533, 1538, 1564, 1586
\@checkend . . . 336, 379, 391, 395
\@chnum . . 1633, 1651, 1681, 1703
\@classi . 1459, 1478, 1533, 1567
\@classii 1459, 1480, 1533, 1569
\@classiii 1459, 1482
\@classiv 1065, 1074, 1161, 1172, 1460, 1484
\@classv . . 884, 974, 1460, 1486, 1534, 1574, 1729
\@classv@LaTeX 884, 1729
\@classv@ltx 974, 1733
\@classvi 1534, 1576
\@classvii 1535, 1578
\@classviii 1535, 1580
\@classx . . 990, 1048, 1536, 1537, 1582, 1584, 1741
\@classx@array 990, 1741
\@classx@array@new . 1048, 1751
\@classz . 1064, 1073, 1160, 1171, 1459, 1476, 1533, 1565
\@ctrerr 735
\@currentlabel 409, 425, 503, 543
\@currentlabelname 32
\@currentlabelname 814
\@dblarg 751, 752
\@dblfloat 650, 652, 671
\@dblfloat@LaTeX 650, 671
\@defaultsubs 360
\@depth . . 1115, 1116, 1140, 1146, 1198, 1242, 1251, 1273, 1428
\@dofilelist 353
\@eha 161
\@enddocumenthook 335
\@endpbox 1204, 1227, 1333, 1342, 1454, 1470, 1524, 1555, 1731, 1738
\@eqcnt . . 412, 417, 418, 420, 426, 439, 442, 445
\@eqncr 40
\@eqncr 414, 428
\@eqnsel 416, 438
\@eqnswtrue 410, 426
\@eqpen 40
\@esphack 545
\@expast 1455, 1471
\@finalstrut 508
\@firstampfalse 1605, 1608, 1631, 1649, 1677, 1701, 1747, 1760
\@firstamptrue 1450, 1465, 1523, 1552
\@firstoftwo 846
\@float 23, 28
\@float@LaTeX 648, 649, 670
\@floatboxreset 618

\@flushglue 405, 1439
\@font@warning 355
\@footnotemark 20, 21
\@footnotemark 483
\@footnotetext 20, 21
\@footnotetext 460, 467, 487, 599
\@gobble@opt@one 56
\@gobble@opt@one 1784
\@halignto 1163, 1174, 1179, 1190, 1202, 1236, 1255, 1300
\@hang@from 780, 811
\@hang@froms 821, 842
\@hangfrom 30–32
\@hangfrom 811
\@hangfrom@section 31
\@height . 1115, 1116, 1197, 1241, 1250, 1272, 1374, 1427, 1434
\@ialph 730, 731
\@if@empty 277, 1877
\@if@sw . . 280, 281, 345, 362, 364
\@ifcat 277
\@ifdim . 277, 354, 776, 817, 1426, 1872, 1879
\@ifempty 14
\@ifeof 277
\@ifhbox 277
\@ifhmode 277
\@ifinner 277, 565, 1357
\@ifl@t@r 261
\@ifmmode . 277, 1167, 1183, 1351, 1353, 1412
\@ifnextchar 460, 463, 1096, 1099, 1102, 1105, 1113, 1114, 1130, 1134
\@ifnotempty 14
\@ifnotrelax 13
\@ifnotrelax 255
\@ifnum . 277, 758, 787, 801, 1773
\@ifodd 277, 1775
\@ifpackageloaded 180, 858
\@ifstar . . 750, 1111, 1112, 1122, 1127
\@ifundefined . 21, 767, 780, 785, 796, 799, 821, 827, 832, 836, 1832, 1838
\@ifvbox 277
\@ifvmode 277
\@ifvoid 277, 562, 573
\@ifx 13, 14
\@ifx 255, 256, 259, 277, 298, 363, 449, 718, 846, 869–889, 983–995, 1058, 1059, 1778
\@ifx@empty 277
\@ifxundefined 13
\@ifxundefined . . . 192, 227, 232, 237, 255, 482, 590, 638, 648
\@ifxundefined@cs 259, 656, 664, 684
\@iiiparbox 584
\@itemlabel 1896
\@lastchclass 1450, 1460, 1461, 1466, 1488, 1490, 1523, 1538, 1539, 1551, 1586, 1588, 1629, 1638, 1666, 1690, 1742, 1752
\@latex@warning@no@line . . 365
\@listdepth . . . 1887, 1890, 1895
\@mainaux 16
\@mainaux 346
\@makecol 9
\@makefnmark 634
\@makefntext 507, 601
\@makeother 711, 715
\@minipagefalse 581
\@mklab 1897
\@mkpream 50
\@mkpream 876, 966, 989, 1047, 1200, 1216, 1254, 1277, 1331, 1339, 1449, 1522
\@mkpream@LaTeX 876, 1449
\@mkpream@array 989, 1522
\@mkpream@array@new 1047, 1549
\@mkpream@ltx 966, 1464
\@mkpream@relax 50
\@mkpream@relax 1514, 1554
\@mpargs 584
\@mpfn 471, 477, 595
\@mpfootins 25
\@mpfootins . 551, 554, 555, 562, 563, 566, 569, 573, 574
\@mpfootnotetext 21
\@mpfootnotetext . 487, 599, 600

\@mpmakefntext 586
\@multiplelabels 363, 371
\@namedef . 677, 1328, 1832, 1838
\@nameuse . 177, 589, 1214, 1290, 1833, 1863
\@ne 11, 59
\@newl@bel 348
\@nextchar 1457, 1458, 1472, 1474, 1530, 1561, 1718, 1721, 1724, 1727, 1730, 1737
\@nmbrlistfalse 1898
\@nobreakfalse 1794
\@normalcr 401
\@nul 716, 727
\@outputdblcol 9
\@parboxrestore 502, 617
\@preamble 1201, 1203, 1211, 1217, 1221, 1236, 1255, 1256, 1266, 1278, 1280, 1284, 1300, 1309, 1314, 1334, 1344, 1448, 1451, 1467, 1523, 1550, 1605, 1611, 1620, 1632, 1679, 1680
\@preamerr 1210, 1234, 1462, 1494, 1496, 1543, 1544, 1546, 1594, 1596, 1600
\@provide 297, 860, 862, 865
\@refundefined 361
\@resetactivechars 219
\@rightskip 404
\@runin@to 31
\@runin@to 796, 812
\@runin@tos 832, 843
\@seccntformat 30, 31
\@seccntformat 768
\@secpenalty 748
\@sect 30, 31
\@sect 755
\@sect@ltx 752, 757
\@setckpt 347
\@sharp . . 1207, 1230, 1262, 1295, 1332, 1340, 1453, 1469, 1504, 1510, 1524, 1544, 1553, 1596, 1634, 1635, 1653, 1657, 1661, 1682, 1684, 1686, 1705, 1709, 1713, 1731, 1738
\@ssect 30, 31, 33
\@ssect 813
\@ssect@ltx 751, 813
\@startcolumn 9
\@startpbox 1204, 1226, 1333, 1342, 1454, 1470, 1520, 1524, 1555, 1730, 1736
\@startsection 30, 33
\@startsection 736
\@startsection@hook . . 737, 754
\@starttoc 57
\@starttoc 1785
\@svsec 31
\@svsec . . 760, 766, 768, 771, 781, 797
\@svsechd 794, 830
\@tabacol . 879, 969, 1050, 1063, 1072, 1089, 1616
\@tabacol@LaTeX 879, 1619
\@tabacol@ltx . . 969, 1050, 1622
\@tabacoll 1070, 1087, 1616
\@tabacolr 1071, 1088, 1625
\@tabarray 39
\@tabarray 872, 962, 985, 1043, 1067, 1076, 1084, 1093, 1095, 1164, 1175, 1180, 1191
\@tabarray@LaTeX 872, 1095
\@tabarray@array 985, 1101
\@tabarray@array@new 1043, 1104
\@tabarray@ltx 962, 1098
\@tabclassiv 881, 971, 1065, 1074, 1717
\@tabclassiv@LaTeX . . 881, 1717
\@tabclassiv@ltx 971, 1720
\@tabclassz 883, 973, 1064, 1073, 1665
\@tabclassz@LaTeX . . . 883, 1665
\@tabclassz@ltx 973, 1689
\@tabular 39
\@tabular . . . 864, 871, 961, 984, 1042, 1060
\@tabular@LaTeX 871, 1060
\@tabular@array 984, 1078
\@tabular@array@new 1042, 1086
\@tabular@ltx 961, 1069

\@tabularcr 40
\@tabularcr 886, 976, 1058, 1066, 1075, 1107
\@tabularcr@LaTeX 40
\@tabularcr@LaTeX . . . 886, 1111
\@tabularcr@ltx 976, 1112
\@tbpen 40
\@tbpen 1107
\@tempa 181, 183, 717, 718
\@tempb 708, 718
\@tempc 706, 724
\@tempcntb 1765, 1768, 1769, 1772, 1774, 1775
\@tempdima 59
\@tempskipa . 740, 742, 743, 748, 775, 776, 816, 817
\@temptokena . 1525, 1531, 1556, 1561
\@testdef 348
\@testpach 1458, 1474, 1532, 1563
\@testtrue 1769, 1775
\@tfor . . . 1456, 1472, 1530, 1561
\@thefnmark 472, 478, 504
\@toodeep 1888
\@totalleftmargin . 1907, 1910, 1914, 1916
\@tpfootnotetext 487
\@trivlist 1900
\@undefined . . . 846, 1090, 1187
\@unexpandable@protect . . 1452, 1468
\@whilesw 1526, 1557
\@width . . 1115, 1116, 1140, 1146, 1199, 1243, 1252, 1274, 1429
\@write@floatline 29
\@write@floatline 714
\@writefile 530
\@xargarraycr 40
\@xargarraycr 888, 978, 994, 1055, 1107
\@xargarraycr@LaTeX . 888, 1115
\@xargarraycr@array . 994, 1137
\@xargarraycr@ltx . . . 978, 1116
\@xargarraycr@new . . 1055, 1143
\@xarraycr 40
\@xarraycr 993, 1054, 1107
\@xarraycr@array 993, 1129
\@xarraycr@new 1054, 1133
\@xbitor 1765
\@xbitor@LaTeX 1765, 1778
\@xbitor@ltx 1771, 1783
\@xfloat 586
\@xfloat@LaTeX 586, 591
\@xfloat@anchored 591, 609
\@xfloat@prep 588, 593
\@xfootnote 459
\@xfootnotemark 20, 21
\@xfootnotemark 459
\@xfootnotemark@ltx . . 469, 485
\@xfootnotenext 26
\@xhline 1375, 1381, 1382
\@xhline@unneeded 1384
\@xifempty 14
\@xsect 809, 840
\@xtabularcr 40
\@xtabularcr . . . 887, 977, 1107
\@xtabularcr@LaTeX . . 887, 1113
\@xtabularcr@ltx 977, 1114
\@yargarraycr 40
\@yargarraycr 889, 979, 995, 1056, 1107
\@yargarraycr@LaTeX . 889, 1117
\@yargarraycr@array . 995, 1150
\@yargarraycr@ltx . . . 979, 1118
\@yargarraycr@new . . 1056, 1154
\@yfootnote 459
\[. 715
\\ 40
\{ 711, 715
\} 711, 715
\] 715
\^ 710, 723
00readme.txt 3, 4

\␣ . . 69, 93, 94, 98, 117, 123, 128, 134, 135, 181

A

\abovedisplayshortskip . . . 434
\abovedisplayskip 429–434
acrofont document class 1

acrofont.sty 1–3
acrofont.sty document class . . 3
\active 710
\addcontentsline 33
\addcontentsline . 532, 786, 800, 828, 837
\addtocontents 524, 533
\AddToHook 303, 377
\aftergroup 46
\aftergroup . . . 1351, 1358, 1867
\appdef 12, 15, 16, 24
\appdef . . 219, 226, 481, 547, 604, 844, 857, 1110, 1221, 1284, 1422, 1446, 1448
\appdef@e 244
\appdef@eval 249
\appdef@val 241, 250
argument
  float 29
  text 34
\array 41
\array . 873, 963, 986, 1044, 1158
array document class 9, 10, 34, 35, 38, 40, 41, 50
array environment 2, 34, 39
\array@array 986, 1177
\array@array@new . . . 1044, 1182
\array@default . 669, 1099, 1105, 1349
\array@hook 863
\array@LaTeX 873, 1158
\array@ltx 963, 1166
\array@row@pre . 1224, 1287, 1365
\array@row@pst . 1306, 1312, 1365
\array@row@rst 46
\array@row@rst 1365, 1509, 1518, 1653, 1657, 1661, 1705, 1709, 1713, 1738
\arraycolsep 417, 418, 1178, 1611, 1614
\arrayrulewidth 1374, 1388, 1434
\arraystretch 1197, 1198, 1241, 1242, 1250, 1251, 1272, 1273, 1426, 1428
\AtBeginDocument 15, 23, 24, 26, 27
\AtBeginDocument 320
\AtEndDocument 15
\AtEndDocument 323
\AtEndOfClass 191
\author 79

B

\baselineskip 12
\baselineskip . 1209, 1232, 1264, 1297
\baselinesklip 12
\begin 15
\begin@float@pagebreak . . . 611, 629, 662, 685
\belowdisplayshortskip . . . 433
\belowdisplayskip 432
\bgroup 21, 46
\body@font 1371, 1406
\botrule 46
\botrule 1370
\box 58

C

\c@mpfootnote 598
\c@secnumdepth . . . 758, 787, 801
\caption 25
\caption 547
\cat@letter 198
\catcode 710, 715
\cell@fil 1401, 1444, 1709, 1713
\cell@font 1400, 1407, 1509, 1519, 1705, 1709, 1713, 1738
\centering 18
\chardef 197–199, 706
\check@aux 18
\check@aux 330, 397, 398
\class@documenthook 15
\class@documenthook . 319, 481, 547, 604, 857
\class@enddocumenthook . 15, 24
\class@enddocumenthook 189, 319
\class@err 161
\class@info . 161, 450, 755, 955, 957, 1036, 1038, 1779, 1781

\class@name 10
\class@name . . 161–163, 182, 192
\class@warn 161
\class@warn@end . . 165, 170, 175, 188
⟨*class customization commands*⟩ placeholder 9
\ClassError 161, 182
classes.dtx 23
classes.dtx document class . . 28
\ClassInfo 163
\classname 85, 92, 140
\ClassWarningNoLine 162
\clear@document 17, 18
\clear@document . . 337, 342, 396, 399
\clearpage 17, 18
\clearpage 342, 392, 399
\closeout 346, 679
\col@sep . 1081, 1090, 1178, 1187
\color@begingroup 506
\color@endgroup 510, 582
\colrule 46
\colrule 1370
\column@fil 1370, 1405
\column@font 1370, 1404
\columnsep 519, 521
\columnwidth 22
\columnwidth 514, 616
\contentsline 24, 56
\contentsline 532
\copy 1412
\count 1765, 1772
\count@ . . 1505, 1511, 1527, 1558
\count@i 224
\count@ii 224
\countdef 224, 225
\crcr 1303, 1306, 1309, 1312, 1317, 1393
\csname 13, 42
\csname 22, 202, 204, 208, 210, 259, 280, 471, 504, 538, 641, 643, 646, 647, 675, 676, 680, 689, 693, 695, 706, 709, 772, 780, 785, 790, 796, 799, 804, 821, 827, 832, 836, 1052, 1791, 1792, 1831, 1834, 1837, 1839, 1843, 1844, 1856–1858, 1867, 1895

D

\d@llarbegin . 1059, 1082, 1091, 1179, 1188, 1189
\d@llarend 1083, 1092, 1179, 1189
\deadcycles 339
\detokenize 381, 382
\dimen 57
\dimen@ . . 1425–1427, 1858, 1859
\dimen@iii 195
\dimendef 195
\displaystyle . 416, 419, 438, 444
\displaywidth 415
\do 1457, 1473, 1531, 1562
\do@check@aux 17, 18
\do@check@aux 330, 393, 398
\do@if@floats 23, 27
\do@if@floats 637
doc 4, 6
\DocInput 8
\document 15
\document 301
document class
acrofont 1
acrofont.sty 3
array 9, 10, 34, 35, 38, 40, 41, 50
classes.dtx 28
ftnright 9
hyperref . . 18, 21, 24, 25, 32, 33, 56, 59
hyperref␣package 2, 24
longtable 10, 50
ltxdoc 1, 5, 8
ltxdocext 1
ltxdocext.sty 3
ltxgrid 10, 16, 21
ltxutil 9
document environment 4, 16
\doublerulesep 1387
\dp . 499, 1115, 1116, 1139, 1145, 1198, 1242, 1251, 1273, 1428

E

\edef 50
\edef 229, 234, 239, 381, 382, 1201, 1605, 1611, 1620, 1632, 1679, 1834
\end 29
\end@float@anchored . 613, 614, 622
\end@float@pagebreak 627, 630, 662
\endarray 44
\endarray 857, 874, 964, 987, 1045, 1302, 1320, 1323, 1326
\endarray@array 987, 1308
\endarray@array@new 1045, 1311
\endarray@hook 861, 862
\endarray@LaTeX 874, 1302
\endarray@ltx 44
\endarray@ltx . . 964, 1305, 1312
\endbatchfile 65
\endcsname 22, 202, 204, 208, 210, 259, 280, 471, 504, 538, 641, 643, 646, 647, 675, 676, 680, 689, 693, 695, 706, 709, 772, 780, 785, 790, 796, 799, 804, 821, 827, 832, 836, 1052, 1791, 1792, 1831, 1834, 1837, 1839, 1843, 1844, 1856–1858, 1867, 1895
\enddocument 17, 18
\enddocument 330
\endgroup 15, 17
\endminipage 577
\endpreamble 44
\endtabular . 857, 875, 965, 988, 1046, 1052, 1316, 1328
endtabular environment 45
\endtabular* 1328
\endtabular@array . . . 988, 1322
\endtabular@array@new . . . 1046, 1325
\endtabular@hook 859, 860
\endtabular@LaTeX . . . 875, 1316
\endtabular@ltx 965, 1319
\endwrite@float 699, 709
\endwrite@floats 701
environment
array 2, 34, 39
document 4, 16
endtabular 45
eqnarray 18, 40
figure 23
figure@write 23, 28
longtable 23, 29
minipage 20, 23
tabular 2, 34, 39, 40
\eqnarray 18
\eqnarray 449, 451
eqnarray environment . . . 18, 40
\eqnarray@fleqn@fixed 407
\eqnarray@LaTeX 407
\eqncolsep . . . 440, 443, 452, 453
\errorstopmode 216
\extrarowheight 1248, 1270

F

\f@ur 197, 1551
\false@sw 270, 276, 281, 293, 298, 892, 895, 898, 901, 904, 907, 910, 913, 916, 919, 922, 925, 928, 931, 934, 937, 940, 943, 946, 949, 952, 998, 1001, 1004, 1007, 1010, 1013, 1016, 1019, 1022, 1025, 1028, 1031, 1034
figure environment 23
figure@write environment 23, 28
\figuresname 23
file
.dtx 1, 4–6
.ins 4
.tfm 1
.vf 1
.vpl 1
00readme.txt 3, 4
acrofont.sty 1–3
classes.dtx 23
doc 4, 6
ltfloat.dtx 20
ltxdocext.dtx 1–3
ltxdocext.ins 3
ltxdocext.pdf 1

ltxdocext.sty 1–3
ltxutil 2, 10, 60
ltxutil.drv 5
ltxutil.ins 5
ltxutil.sty 5
makeindex 3
nameref.sty 32
pdfmark.def 32
zpsynocmrv 1
zptmnocmr 1
zptmnocmrm 1
zpzcnocmry 1
\file 17, 47, 48, 103, 106, 112, 113, 122, 128, 131, 132, 134, 137, 138
float, argument 29
\float@end@tag 29
\float@end@tag 714
\floatingpenalty 21
\floatingpenalty 500
\floats@sw 26, 27
\floats@sw . . 586, 590, 591, 607, 608, 638, 639, 666
\flushing 400
\fmtversion 261
\font@submax 354, 357
\fontsubfuzz 354
\footins 488, 574
\footnote 20, 23
\footnote 459, 594, 605
\footnote@latex 594, 605
\footnotemark 20
\footnotemark 459
\footnotesep 498, 508
\footnotesize 496, 1800
\fp@proc@H 608
\fp@proc@h 607
\from 47, 49
\frstrut 1370, 1371, 1408
ftnright document class 9
\fullinterlineskip . . . 223, 564
\futurelet 1374, 1381

G

\gappdef 189, 226
\generate 46
\GetFileInfo 26
\glet@environment 176, 200
\glossary 526

H

\H@@footnotemark . 474, 479, 482, 483
\H@@footnotetext 600
\H@refstepcounter 31
\H@refstepcounter 762, 855
\H@svsec . 759, 763, 781, 797, 815, 822, 823, 833, 834
\halign 44, 50
\halignt@ 196, 437
\hangindent 59
\hb@xt@ 420, 445, 634, 1865, 1875
\hbox 59
\hline 47
\hline 885, 975, 1373
\hline@LaTeX 885, 1373
\hline@ltx 975, 1377
\hline@rule 47
\hline@rule . . 1372, 1380, 1395, 1410
⟨*hooks and bookkeeping*⟩ placeholder 17
\hrule 12
\hsize 15
\ht 1197, 1241, 1247, 1269, 1425, 1426
\Hy@raisedlink 849
\hyper@anchor 847
\hyper@anchorend 33
\hyper@anchorend 851
\hyper@anchorstart 33
\hyper@anchorstart 850
\hyper@last 33
\hyper@last 848
\hyper@linkend 33
\hyper@linkend 853
\hyper@linkstart 33
\hyper@linkstart 852
\hyperanchor 33
hyperref document class . 18, 21, 24, 25, 32, 33, 56, 59

hyperref␣package document class 2, 24

I

\ialign 42
\ialign 1202, 1255
\if 42
\if 267, 279, 1194, 1260
\if@filesw . 345, 362, 1790, 1819
\if@firstamp 1605, 1608
\if@nobreak 745
\if@noskipsec 738
\ifeof 283
\iffalse 1121, 1126
\IfFormatAtLeastTF . . . 260, 263
\ifhbox 284
\ifinner 286
\ifmmode 287
\ifodd 289, 1769
\ifToplevel 52
\ifvbox 290
\ifvoid 292
\ignorespaces 508, 1334, 1347, 1504, 1510, 1682, 1684, 1686, 1705, 1709, 1713, 1730, 1738, 1864, 1871, 1883, 1903
\immediate . . 346, 644, 678, 724, 1792, 1842
\incompatible@package 179
\index 526
\init@documenthook . . 304, 312, 319, 844
\insert 488, 574
\insert@column . 991, 1051, 1502
\insert@column@array 991, 1502
\insert@column@array@new 1051, 1507
\interdisplaylinepenalty . . 40
\interfootnotelinepenalty 497
\interlinepenalty 497, 778, 819
\intertabularlinepenalty . . 40
\intertabularlinepenalty 1107, 1108, 1110, 1112, 1127
\intextsep 629, 630
\item 126, 136, 139
\itemindent 1894

J

\j@nk 299

K

\keepsilent 45

L

\l@ 57
\l@@sections 57
\l@@sections 1848
\l@section 1802
\label 526, 540
\lastbox 567, 683
\LaTeX . . 69, 93, 94, 104, 114, 117, 128, 134, 135, 142
\LaTeXe 98, 123, 153
\leftmargin . . . 1907, 1909, 1914
\leftmargini 1802
\leftmarginii 1803
\leftmarginiii 1804
\leftmariniv 1805
\leftmarginv 1806
\leftmarginvi 1807
\leftskip 59
\leftskip . 402, 1857, 1859, 1916
\let 57
\let@environment 11
\let@environment . 200, 654, 655, 657, 673, 674
\lineskip 1209, 1232, 1263, 1297
\linewidth 437, 515, 522, 1908–1910
\list 59
\list 1886
\listparindent . 1893, 1906, 1913
longtable document class 10, 50
longtable environment . . 23, 29
\loop 14
\loopuntil 295
\loopwhile 295
\lrstrut 1371, 1372, 1409
\LT@array 10
ltfloat.dtx 20
ltxdoc document class . . . 1, 5, 8

ltxdocext document class 1
ltxdocext.dtx 1–3
ltxdocext.ins 3
ltxdocext.pdf 1
ltxdocext.sty 1–3
ltxdocext.sty document class . 3
ltxgrid document class 10, 16, 21
ltxutil 2, 10, 60
ltxutil document class 9
ltxutil.drv 5
ltxutil.ins 5
ltxutil.sty 5

M

\make@footnotetext 21
\make@footnotetext 487
\makeatletter 20, 350, 1788
\makeatother 23
makeindex 3
\makelabel 1897
\maketitle 21
\maketitle 83
\mark 24
\mathindent 427, 457
\meaning 221
\MessageBreak 356
minipage environment . . . 20, 23
\minipagefootnote@drop 21
\minipagefootnote@drop 511, 550
\minipagefootnote@foot . . . 550
\minipagefootnote@here 25
\minipagefootnote@here . . . 548, 550, 623
\minipagefootnote@init 550, 619
\minipagefootnote@pick 492, 550
\Msg 53–59, 61–63
\multicolumn . . . 870, 960, 1329
\multicolumn@LaTeX . . 870, 1329
\multicolumn@ltx 960, 1336
\multispan 1330, 1337

N

nameref.sty 32
\NC@list 1526, 1557
\newbox 1421
\newcount 1107, 1109
\newenvironment 29
\newlabel 24
\newlabel 543
\newlinechar 723
\newwrite 640, 1791
\noalign . 1114, 1116–1118, 1135, 1148, 1152, 1156, 1372, 1374, 1378, 1382, 1394
\noexpand . . 221, 222, 1202, 1255
\normalsize 306, 314
\numberline 59
\numberline@@sections . . . 1862, 1874

O

\obeylines 720
\obsolete@command 164
\oneapage 662
\openout 645, 1792
\other 199

P

\p@equation 409, 425
\package@name 151, 152
\PackageInfo 152
\pagegrid@col 620
\parindent 59
\parindent 632, 1859, 1875, 1906, 1913
\parsep 1901
\parshape 59
\parshape 1910
\parskip 431, 1901
\partopsep 430
\pbox@hook 1445, 1738
pdfmark.def 32
\pdfstringdefDisableCommands 16
\phantomsection 33
\phantomsection . . 759, 764, 815, 845
placeholder
⟨*class customization commands*⟩ 9
⟨*hooks and bookkeeping*⟩ . . 17

⟨*read main .aux and final checks*⟩ 17
⟨*your document here*⟩ 9
\preamble 31
\prepdef 12
\prepdef 226, 310, 548, 675, 676, 768, 770, 859, 861, 864, 1217, 1280
\prepnext@tok 1529, 1560
\prevdepth 12
\prevdepth 223
\print@float 28
\print@float 661
\printfigures 24
\protected 379
\protected@edef 503, 766
\protected@write 525, 542
\protected@xdef 478
\providecommand . . 845, 847–853, 863
\ProvidesFile 4, 6

R

\raggedleft 18
\raggedright 18
⟨*read main .aux and final checks*⟩ placeholder 17
\realfootnote 21
\refstepcounter 31
\relax 31, 34
\repeat 14
\replace@command 169
\replace@environment 174
\RequirePackage 13, 15, 22
\reserved@a . . 1375, 1381, 1385, 1386, 1457, 1472
\reset@font 496
\romannumeral 1895
\rule 508
\rvtx@enddocument@patch . . 376
\rvtx@enddocument@patch@end 385, 392
\rvtx@enddocument@patch@more 387, 390
\rvtx@ifformat@geq 13
\rvtx@ifformat@geq . . 260, 301, 330, 376

S

\samepage 40
\samepage 602, 1110
\save@decl 1534, 1571
\say 12
\say 221, 1385
\saythe 12
\saythe 221
\sc 18, 114
\sec@upcase 34
\sec@upcase 856
\section 110, 686
\Sectionformat 32
\Sectionformat . . . 822, 833, 846
\set@arstrutbox 47
\set@arstrutbox 1215, 1239, 1410
\set@eqnarray@skips 458
\set@footnotewidth 22
\set@footnotewidth 487
\set@footnotewidth@ii 22
\set@footnotewidth@ii 517
\set@listindent 1902, 1918
\set@listindent@ 1912
\set@listindent@parshape 1905, 1918
\set@pica@hook 16
\set@pica@hook . . . 307, 315, 328
\set@tocdim@pagenum 1855, 1870
\set@typeset@protect . . . 1208, 1231, 1296, 1332, 1341
\set@typesize@hook 16
\set@typesize@hook 305, 313, 328
\shipout 12
\showboxbreadth 216
\showboxdepth 216
\showoutput 219
\skip@ 1861, 1864
\splitmaxdepth 499
\splittopskip 498
\stepcounter 408, 425, 477
\stop 183
\StopEventually 6

\string . . 165, 170, 181, 527–530, 543, 756, 1779, 1781, 1843
\strutbox 499, 508, 1197, 1198, 1241, 1242, 1247, 1251, 1269, 1273, 1426, 1428
\subsection 120
\switch@array 34, 40
\switch@array 858, 867
\switch@tabular 34, 40
\switch@tabular 858, 867

T

\t@ 194, 196, 415
\tab@rule 47
\tab@rule 1370, 1371, 1392
\tabcolsep 1081, 1441–1443, 1620
\tableft@skip . . 1233, 1298, 1438
\tableftsep . . . 1441, 1515, 1617
\tableofcontents 108
\tabmid@skip . . 1218, 1281, 1439
\tabmidsep 1442, 1516, 1623
\tabright@skip . 1222, 1285, 1440
\tabrightsep . . 1443, 1517, 1626
\tabskip 42
\tabskip 413, 416, 419, 421, 427, 438, 444, 446, 1203, 1218, 1222, 1233, 1257, 1281, 1285, 1298
\tabular 34
tabular environment 2, 34, 39, 40
\tabular@hook 864, 865
\tabularnewline 1205, 1228, 1266, 1293
\TeX 98
text, argument 34
\texttt 80
\textwidth 518
\thanks 70
\the@toks 1503, 1505, 1508, 1511, 1528, 1547, 1559, 1602
\theequation 409, 425
\thefootnote 597
\thempfn 472, 478, 596
\thempfootnote 596, 597
\thepage 534, 543
\thr@@ 11
\thr@@ . 195, 420, 445, 1543, 1594
\title 68
\toc@@font 56
\toc@@font 1800, 1809
\toc@letdimen . . 1815, 1816, 1836
\toc@post 57
\toc@post 1795, 1799
\toc@post@auto 57
\toc@post@auto 1808
\toc@pre 57
\toc@pre 1787, 1798
\toc@pre@auto 57
\toc@pre@auto 1808
\toc@setindent . 1811–1814, 1830
\toc@writedimen 1821–1826, 1841
\tocdim@appendix 1806
\tocdim@min . . . 1831, 1837, 1885
\tocdim@pagenum 1807, 1872
\tocdim@paragraph 1805
\tocdim@section 1802
\tocdim@subsection 1803
\tocdim@subsubsection . . . 1804
\tocleft@pagenum 1860
\tocleft@section 58
\tocmax@section 58
\toks 1547, 1602
\toks@ . . . 227, 229, 232, 234, 237, 239, 707, 724, 726
\toks@ii 12
\toks@ii . 228, 229, 233, 234, 238, 239, 254
\toksdef 254
\toprule 46
\toprule 1370
\topsep 429
\traceoutput 12
\traceoutput 218
\tracingall 12
\tracingcommands 213
\tracinglostchars 214
\tracingmacros 215
\tracingonline 213
\tracingoutput 12
\tracingoutput 214
\tracingpages 214

\tracingparagraphs 215
\tracingplain 12
\tracingplain 212
\tracingrestores 215
\tracingstats 213
\trigger@float@par . . 672, 675, 676, 682
\triggerpar 661, 672
\true@sw . 268, 275, 281, 293, 298, 316, 890, 996
\tw@ 11

U

\unhbox 1865, 1880
\unhcopy 1412
\unrestored@protected@xdef 472
\unskip 579, 1138, 1144, 1371, 1372, 1504, 1510, 1682, 1684, 1686, 1705, 1709, 1713, 1864
\unvbox 555, 569, 574, 584
\unvcopy 566
\uppercase 34
\url 14, 89, 95

V

\vbox . . 554, 612, 1194, 1260, 1350
\vcenter 46
\vcenter . 1194, 1260, 1351, 1355, 1358
\voidb@x 551
\vrule 1115, 1116, 1140, 1146, 1197, 1241, 1249, 1272, 1427
\vsize 15
\vtop 566, 1194, 1260, 1349

W

\write 724, 1842
\write@@float 698, 702
\write@float 23, 28, 29
\write@float 651, 698
\write@floatline 29
\write@floatline 712, 714
\write@floats 652, 698

X

\x 381, 384
\xdef 1255, 1278

Y

\y 382, 384
⟨*your document here*⟩ placeholder 9

Z

\z@ 58
\z@skip . . 402–404, 416, 421, 438, 446, 624, 1203, 1209, 1232, 1438–1440, 1865
zpsynocmrv 1
zptmnocmr 1
zptmnocmrm 1
zpzcnocmry 1